Module-1
TCP/IP Protocol Suite

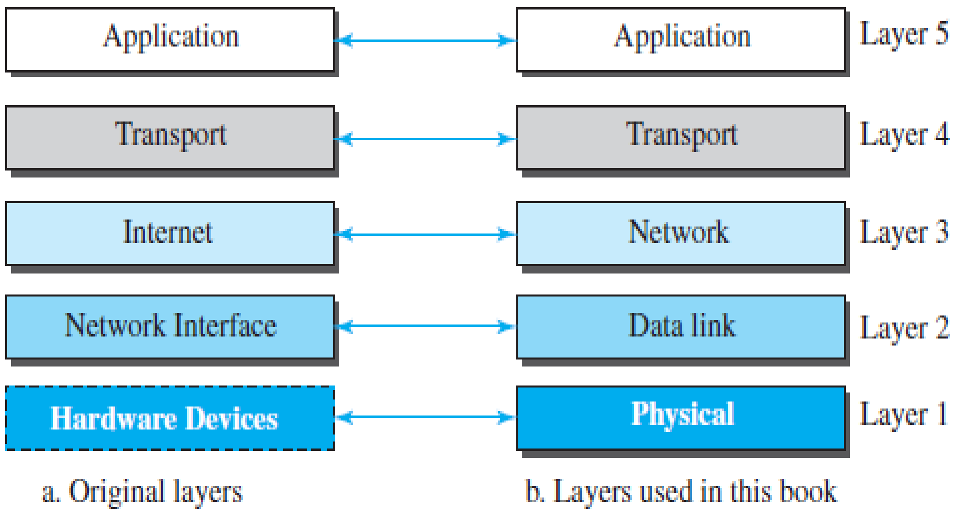
TCP/IP PROTOCOL SUITE

(Transmission Control Protocol / Internet Protocol)

- TCP/IP is a protocol suite (a set of protocols organized in different layers) used in the Internet today. It is a network model designed to support network communication.

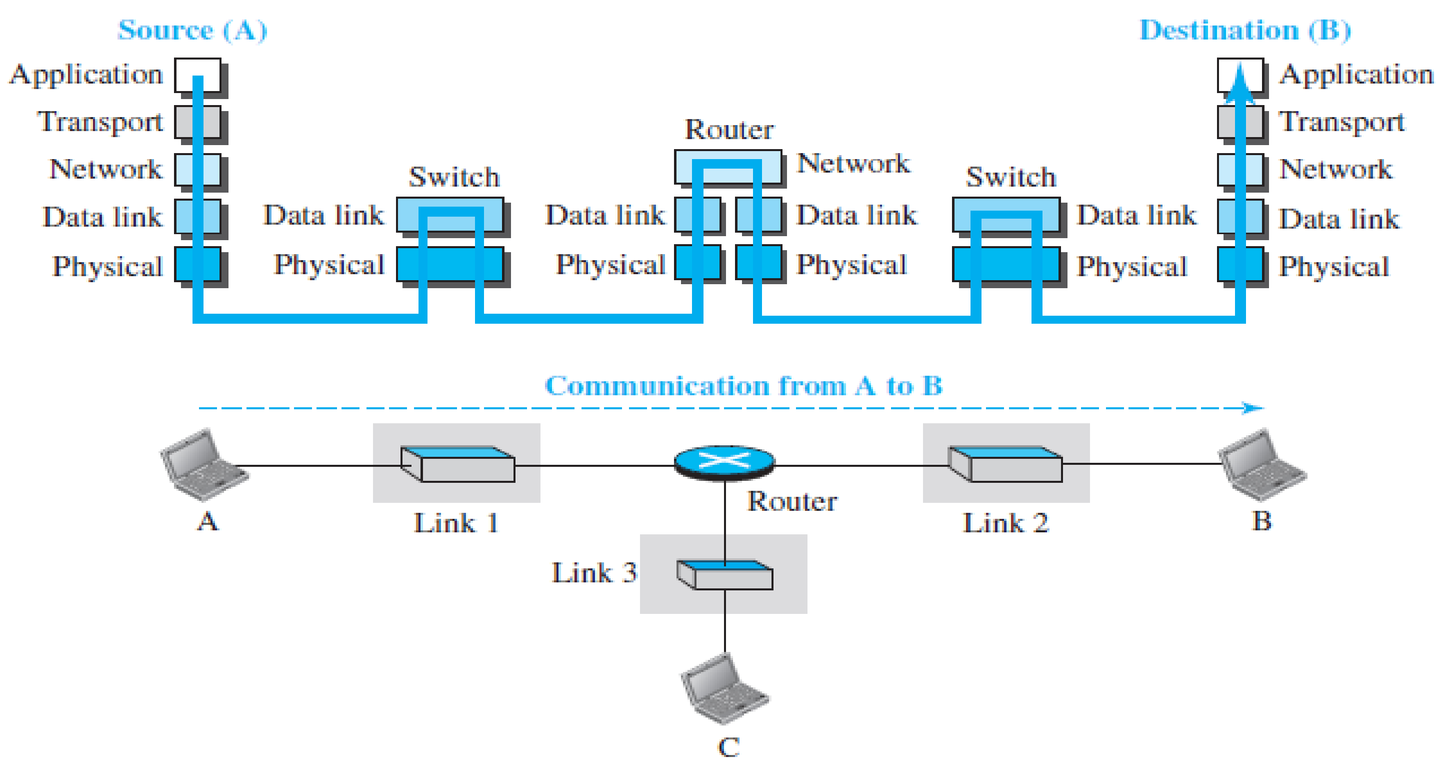
- The original TCP/IP protocol suite was defined as four software layers built upon the hardware. Today, however, TCP/IP is thought of as a **five-layer** model.

Figure 2.4 *Layers in the TCP/IP protocol suite*



Layered Architecture

Figure 2.5 *Communication through an internet*



We want to use the suite in a small internet made up of three LANs (links), each with a link-layer switch. We also assume that the links are connected by one router.

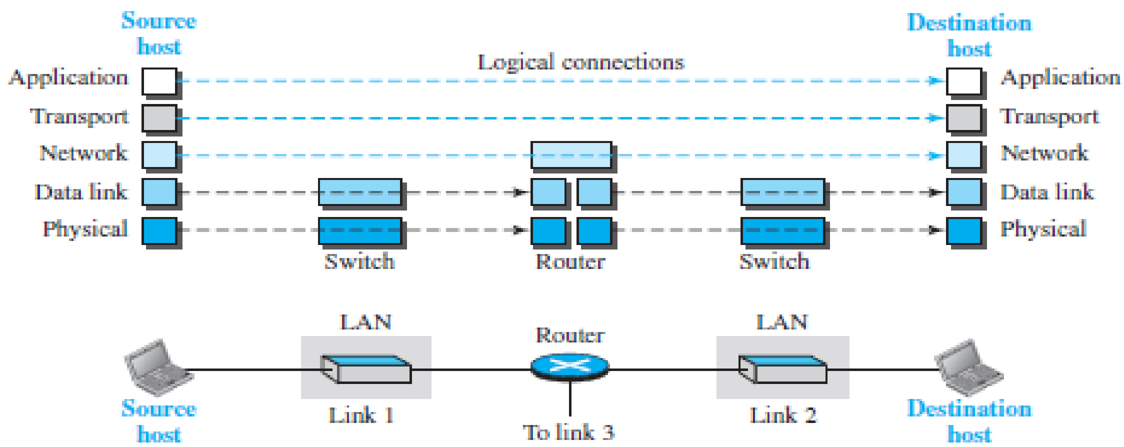
Each device is involved with a set of layers depending on the role of the device in the internet

Layered Architecture

- Let us assume that computer A communicates with computer B.
- Five communicating devices in this communication: Source host (computer A), the link-layer switch in link 1, the router, the link-layer switch in link 2, and the destination host (computer B).
- Each device is involved with a set of layers depending on the role of the device in the internet.
- The two hosts are involved in all five layers:
  - The source host needs to create a message in the application layer and send it down the layers.
  - The destination host needs to receive the communication at the physical layer and then deliver it through the other layers to the application layer.
- The router is involved in only three layers - router is used only for routing

Layers in the TCP/IP Protocol Suite

Figure 2.6 *Logical connections between layers of the TCP/IP protocol suite*



- The duty of the application, transport, and network layers is end-to-end.
- The duty of the data-link and physical layers is hop-to-hop, in which a hop is a host or router.
- The domain of duty of the top three layers is the internet, and the domain of duty of the two lower layers is the link.
- In the top three layers, the data unit (packets) should not be changed by any router or link-layer switch. In the bottom two layers, the packet created by the host is changed only by the routers, not by the link-layer switches.

Layered Architecture

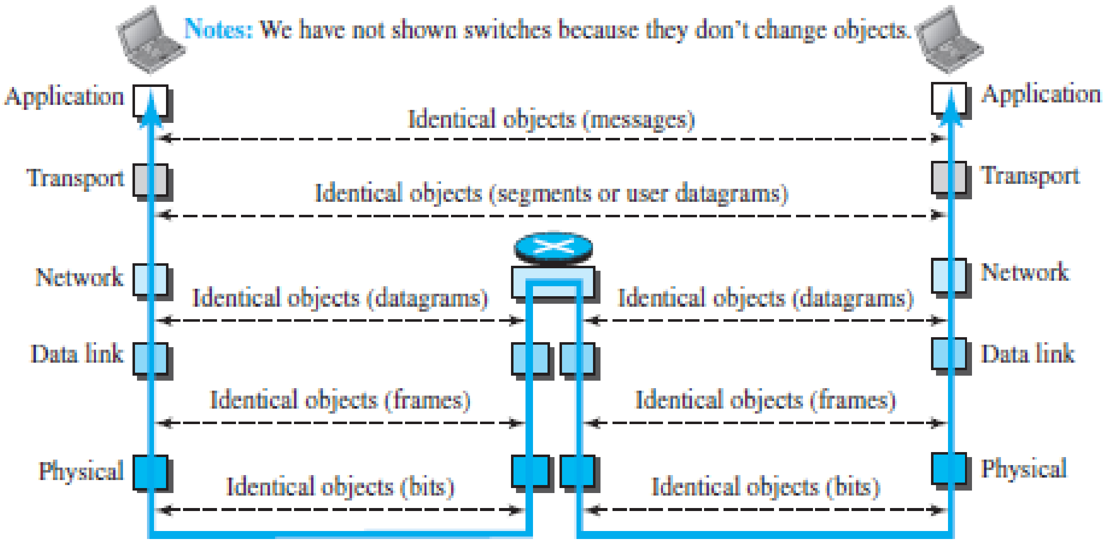
- Although a router is always involved in one network layer, it is involved in *n* combinations of link and physical layers in which *n* is the number of links the router is connected to.

- Each link may be using different link-layer and physical-layer protocols; the router needs to receive a packet from link 1 based on one pair of protocols and deliver it to link 2 based on another pair of protocols.

- Although each switch in the above figure has two different connections, the connections are in the same link, which uses only one set of protocols

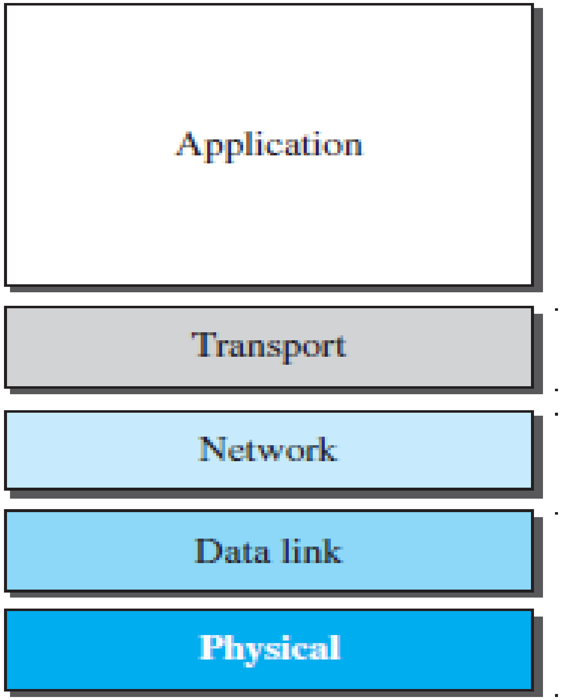
Layers in the TCP/IP Protocol Suite

Figure 2.7 *Identical objects in the TCP/IP protocol suite*



- The identical objects below each layer related to each device
- The logical connection at the network layer is between the two hosts, we can only say that identical objects exist between two hops in this case because a router may fragment the packet at the network layer and send more packets than received.

Description of Each Layer



TCP/IP Protocol Suite

Application Layer:

- Layer with which user interacts
- Responsible for providing services to the user.
- Used by user applications (google chrome, firefox, Microsoft outlook) that pass message from one computer to another in a network. Applications use application protocols to perform their activities.
- Communication at the application layer is between two processes (two programs running at this layer).

Protocols:

- Hypertext Transfer Protocol (HTTP) - for accessing the World Wide Web (WWW).
- The Simple Mail Transfer Protocol (SMTP) -used in electronic mail (e-mail) service.
- The File Transfer Protocol (FTP) -used for transferring files from one host to another.
- The Terminal Network (TELNET) and Secure Shell (SSH) are used for accessing a site remotely.
- The Simple Network Management Protocol (SNMP) is used by an administrator to manage the Internet at global and local levels.
- The Domain Name System (DNS) is used by other protocols to find the network-layer address of a computer.

Transport Layer

- Service-point addressing (Port address)
- Segmentation and reassembly (Sequence number)
- Connection control (Connectionless (UDP) or connection oriented (TCP))
- Flow control (end to end)
- Error Control (Process to Process)

Protocols: TCP, UDP, SCTP

Network Layer

- Routing
- Path determination
- Logical addressing
- IP is a connectionless protocol (unreliable) that provides no flow control, no error control, and no congestion control services.

Protocols: IP, ICMP (ping uses), IGMP (multicasting)

Data Link Layer

- Framing (stream of bits into manageable data units)
- Physical addressing (MAC Address)
- Flow Control (mechanism for overwhelming the receiver)
- Error Control (trailer, retransmission)
- Access Control (defining master device in the same link)
- Error detection and correction
- MAC (data encapsulation, access medium) and LLC (flow control, error control) are the sublayers of data link layer

Protocols: Ethernet, 802.11,CSMA/CD,CSMA/CA, token passing

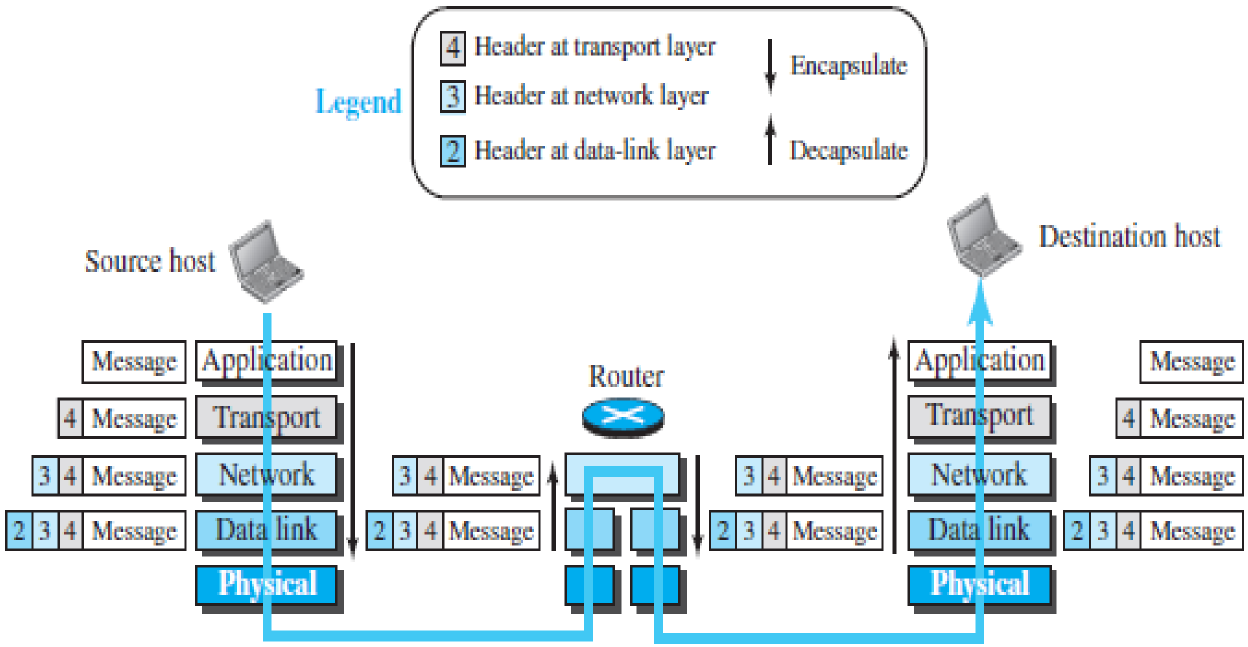
Physical Layer

- Responsible for movements of individual bits from one hop (node) to the next
- The purpose of the Physical layer is to create the electrical, optical, or microwave signal that represents the bits

Protocols: Bluetooth, Ethernet (802.3), DSL, ISDN, 10 Base T

Encapsulation and Decapsulation

Figure 2.8 *Encapsulation/Decapsulation*



Encapsulation: Adding a chunk of data (header) at each layer as data goes from Application to physical layer.

Decapsulation: Reading/removing the data (header) at each layer as data goes from physical to application layer

Encapsulation at source

Encapsulation at source:

- At the **application layer**, the data to be exchanged is referred to as a **message**. A message normally does not contain any header or trailer.

- The **transport layer** adds the **transport layer header** to the payload,
  - contains the identifiers (port numbers) of the source and destination application programs that want to communicate
  - information that is needed for the end-to end delivery of the message, such as information needed for flow, error control, or congestion control.

- The **network layer** takes the transport-layer packet as data or payload and adds its own **header** to the payload.
  - The header contains the addresses of the source and destination hosts (IP address)
  - some more information used for error checking of the header, fragmentation information

- The **data-link layer** takes the network-layer packet as data or payload and adds its own header, which contains
  - the link-layer addresses of the host or the next hop (the router). The result is the link-layer packet, which is called a frame.

- The frame is passed to the **physical layer** for transmission.

Decapsulation and Encapsulation at the Router

- After the set of bits are delivered to the data-link layer, this layer decapsulates the datagram from the frame and passes it to the network layer.

- The network layer only inspects the source and destination addresses in the datagram header and consults its forwarding table to find the next hop to which the datagram is to be delivered.

- The contents of the datagram should not be changed by the network layer in the router unless there is a need to fragment the datagram if it is too big to be passed through the next link.

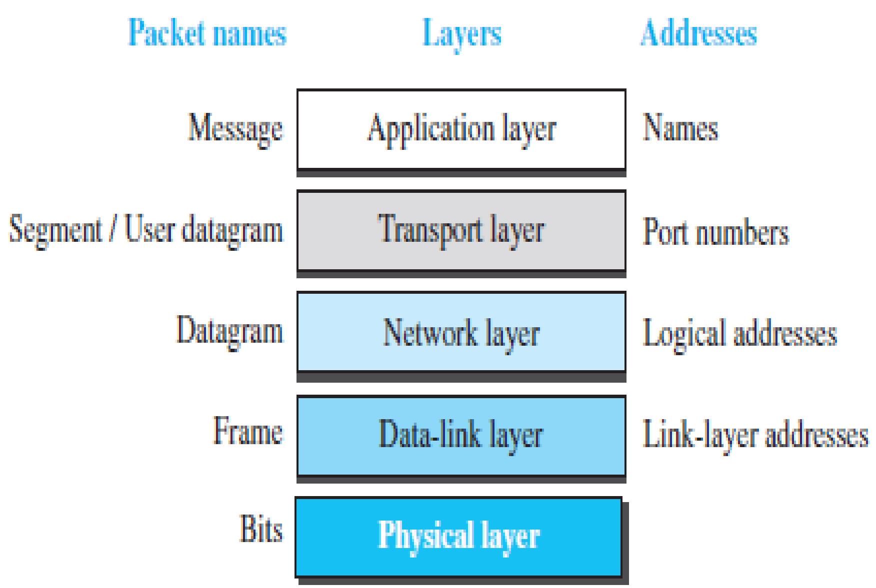
- The data-link layer of the next link encapsulates the datagram in a frame and passes it to the physical layer for transmission

Decapsulation at the Destination Host

- At the destination host, each layer only decapsulates the packet received, removes the payload, and delivers the payload to the next-higher layer protocol until the message reaches the application layer.

Addressing

Figure 2.9 *Addressing in the TCP/IP protocol suite*



Addressing

- At the application layer, we normally use **names** to define the site that provides services, such as *someorg.com*, or the e-mail address, such as *somebody@coldmail.com*.

- At the transport layer, addresses are called **port numbers**, and these define the application-layer programs at the source and destination. Port numbers are local addresses that **distinguish between several programs** running at the same time.

- A network-layer address uniquely defines the connection of a device to the Internet **(logical address**).

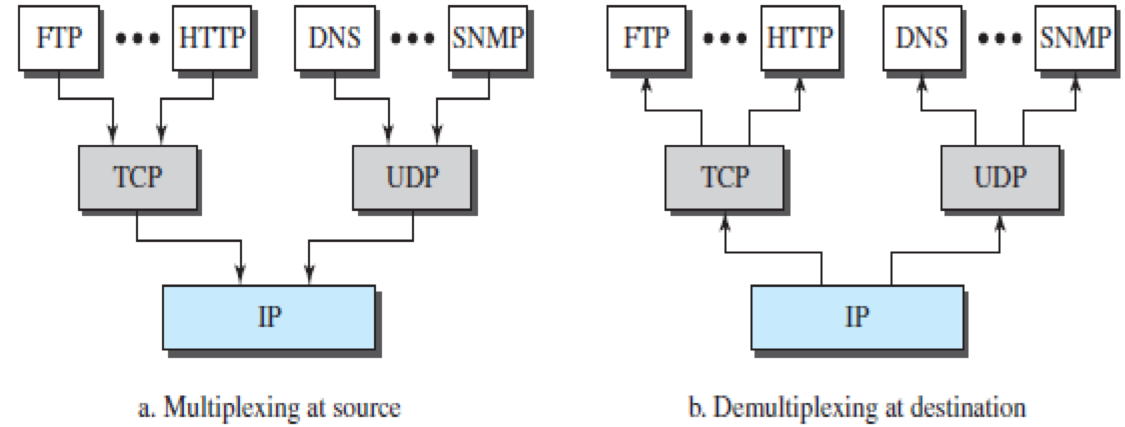
- The link-layer addresses, sometimes called MAC addresses **(physical address)** , are locally defined addresses, each of which defines a specific host or router in a network (LAN or WAN)

Port Numbers -example

Port Number	Protocol
20, 21	File Transfer Protocol (FTP)
22	Secure Shell (SSH)
23	Telnet Protocol
25	Simple Mail Transfer Protocol (SMTP)
53	Domain Name System (DNS)
67, 68	Dynamic Host Configuration Protocol (DHCP)
80	HyperText Transfer Protocol (HTTP)
110	Post Office Protocol (POP3)
137	NetBIOS Name Service
143	Internet Message Access Protocol (IMAP4)
443	Secure HTTP (HTTPS)
445	Microsoft-DS (Active Directory)

Multiplexing and Demultiplexing

Figure 2.10 *Multiplexing and demultiplexing*



- Multiplexing in this case means that a protocol at a layer can encapsulate a packet from several next-higher layer protocols (one at a time).
- Demultiplexing means that a protocol can decapsulate and deliver a packet to several next-higher layer protocols.

Multiplexing and Demultiplexing

- To be able to multiplex and demultiplex, a protocol needs to have a **field in its header** to identify to which protocol the encapsulated packets belong.
- At the transport layer, either UDP or TCP can accept a message from several application-layer protocols.
- At the network layer, IP can accept a segment from TCP or a user datagram from UDP.
- IP can also accept a packet from other protocols such as ICMP, IGMP, and At the data-link layer, a frame may carry the payload coming from IP or other protocols such as ARP so on.

DATA RATE LIMITS

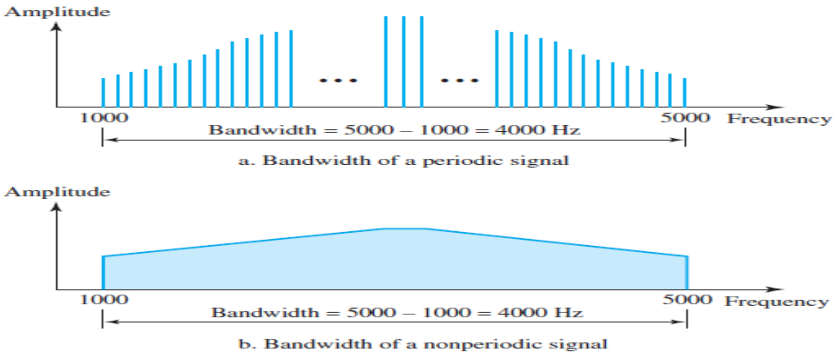
- A very important consideration in data communications is ***how fast we can send data***, in bits per second, over a channel.

- Data rate depends on three factors:
  1. The **bandwidth** available
  2. The **level** of the signals we use
  3. The **quality** of the channel (the level of noise)

- Two theoretical formulas were developed to calculate the data rate:
  1. Nyquist for a noiseless channel,
  2. Shannon for a noisy channel.

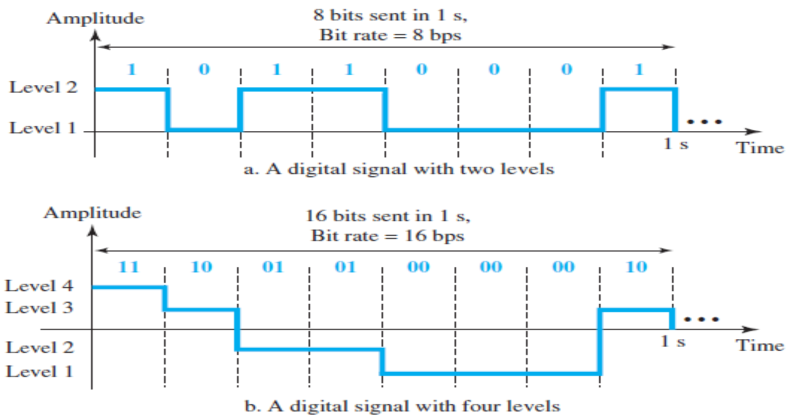
DATA RATE LIMITS

Bandwidth: The bandwidth of a signal is the **difference** between the **highest** and the **lowest frequencies** contained in that signal.



Signal Levels:

Figure 3.17 *Two digital signals: one with two signal levels and the other with four signal levels*



In general, if a signal has ***L* levels**, each level needs **$\log_2$ *L* bits**. $\log_2 4 = 2$ bits

Noiseless Channel: Nyquist Bit Rate

Bitrate: the rate at which bits are transferred from one location to another. It measures how much data is transmitted in a given amount of time.

Nyquist Bit rate

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

- Bandwidth - bandwidth of the channel
- L - number of signal levels used to represent data
- BitRate - the bit rate in bits per second

According to the formula, we might think that, given a specific bandwidth, we can have any bit rate we want by increasing the number of signal levels.

Nyquist Bit Rate

1) Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

2) Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 4 = 12{,}000 \text{ bps}$$

3) We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

Solution

We can use the Nyquist formula as shown:

$$265{,}000 = 2 \times 20{,}000 \times \log_2 L \longrightarrow \log_2 L = 6.625 \longrightarrow L = 2^{6.625} = 98.7 \text{ levels}$$

Tradeoff

- Tradeoffs:
  - Increase the bandwidth, increases the data rate
  - Increase the signal levels, increases the data rate
  - Increase the signal levels, harder for receiver to interpret the bits (practical limit to M)

- When we increase the number of signal levels, we impose a burden on the receiver.

- If the number of levels in a signal is just 2, the receiver can easily distinguish between a 0 and a 1.

- If the level of a signal is 64, the receiver must be very sophisticated to distinguish between 64 different levels.

Noisy Channel: Shannon Capacity

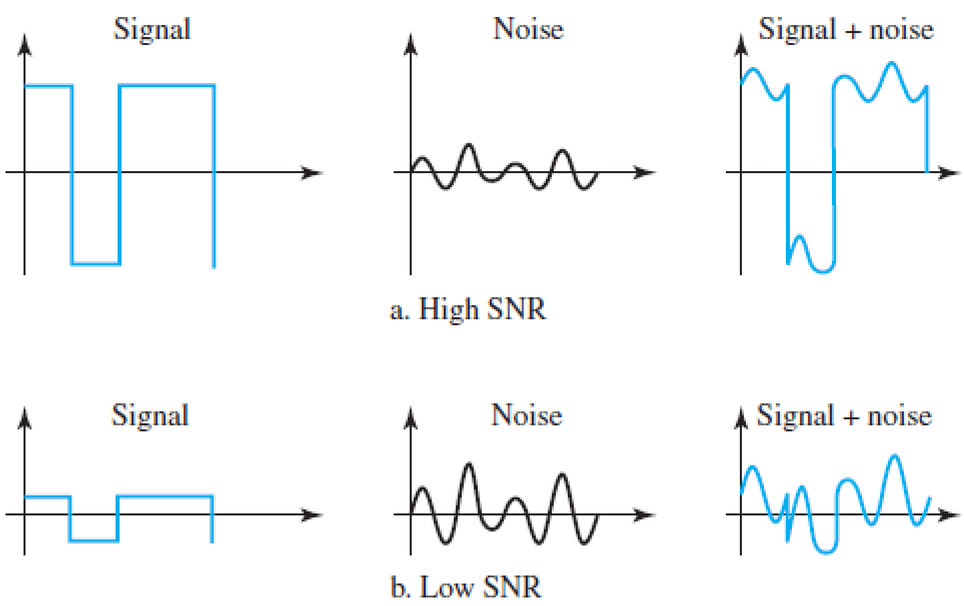
- We cannot have a noiseless channel; the channel is always noisy
- In 1944, Claude Shannon introduced a formula, called the **Shannon capacity**, to determine the theoretical highest data rate for a noisy channel:

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$$

- Bandwidth is the bandwidth of the channel
- SNR is the signal-to noise ratio
- Capacity is the capacity of the channel in bits per second
- No indication of the signal level, which means that no matter how many levels we have, we cannot achieve a data rate higher than the capacity of the channel.

SNR (Signal to Noise Ratio)

$$SNR = \frac{\text{average signal power}}{\text{average noise power}}$$



Noisy Channel: Shannon Capacity

1) Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as

$$C = B \log_2 (1 + \text{SNR}) = B \log_2(1 + 0) = B \log_2 1 = B \times 0 = 0$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

2) We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as

$$C = B \log_2 (1 + \text{SNR}) = 3000 \log_2(1 + 3162) = 3000 \times 11.62 = 34{,}860 \text{ bps}$$

This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

Example 3.39

The signal-to-noise ratio is often given in decibels. Assume that $SNR_{dB} = 36$ and the channel bandwidth is 2 MHz. The theoretical channel capacity can be calculated as

$$SNR_{dB} = 10 \log_{10} SNR \longrightarrow SNR = 10^{SNR_{dB}/10} \longrightarrow SNR = 10^{3.6} = 3981$$

$$C = B \log_2(1 + SNR) = 2 \times 10^6 \times \log_2 3982 = 24 \text{ Mbps}$$

Using Both Limits

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution

First, we use the Shannon formula to find the upper limit.

$$C = B \log_2(1 + \text{SNR}) = 10^6 \log_2(1 + 63) = 10^6 \log_2 64 = 6 \text{ Mbps}$$

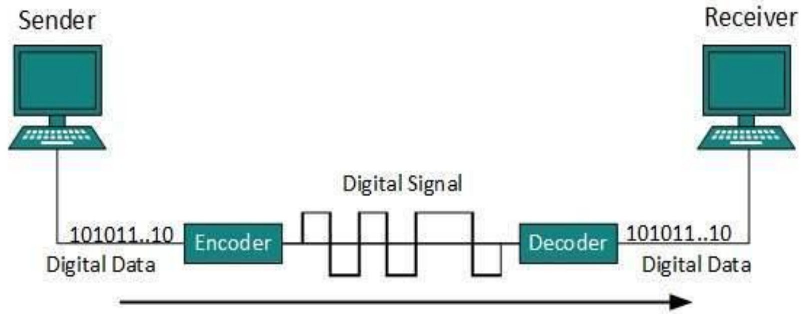
The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps, for example. Then we use the Nyquist formula to find the number of signal levels.

$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \longrightarrow L = 4$$

Note

- **The Shannon capacity gives us the upper limit;** The equation sets the theoretical upper limit on data rate, which of course is not fully achieved in practice
- **The Nyquist formula tells us how many signal levels we need**

Digital-to-Digital conversion



Line coding | **Block coding** | **Scrambling**

Always needed

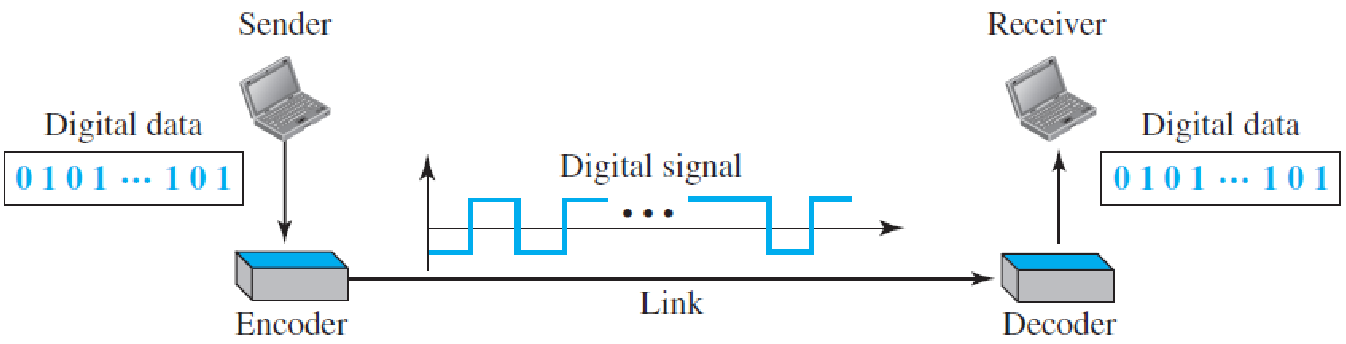
may or may not be needed

Line Coding

- Line coding converts a sequence of digital bits to a digital signal.

- At the sender, digital data are encoded into a digital signal; at the receiver, the digital data are recreated by decoding the digital signal.

Figure 4.1 *Line coding and decoding*



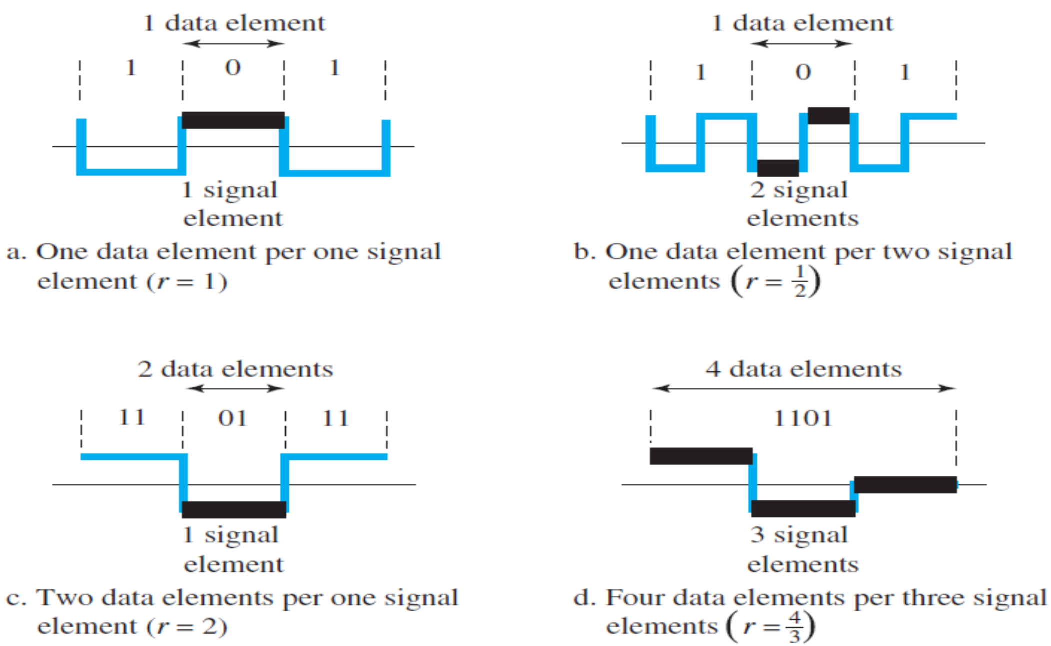
Characteristics of line coding schemes/ Factors to consider in digital signaling

- **Data Element Versus Signal Element**

A data element is the smallest entity that can represent a piece of information: this is the bit	A signal element carries data elements.
Data elements are what we need to send	Signal elements are what we can send.
Data elements are being carried	Signal elements are the carriers

ratio ***r*** - which is the number of data elements carried by each signal element

Figure 4.2 *Signal element versus data element*



Characteristics of line coding schemes

- **Data Rate Versus Signal Rate**

The data rate defines the number of data elements (bits) sent in 1s	The signal rate is the number of signal elements sent in 1s.
The unit is bits per second (bps).	The unit is the baud.
The data rate is sometimes called the bit rate	The signal rate is sometimes called the pulse rate, the modulation rate, or the baud rate.

"One goal in data communications is to increase the data rate while decreasing the signal rate"

- Increasing the data rate increases the speed of transmission; decreasing the signal rate decreases the bandwidth requirement.

- In our vehicle-people analogy, we need to carry more people in fewer vehicles to prevent traffic jams. We have a limited bandwidth in our transportation system.

Characteristics of line coding schemes

Data Rate Versus Signal Rate

We now need to consider the relationship between data rate (*N*) and signal rate (*S*)

$$S = N/r$$

This relationship, of course, depends on the value of *r*. It also depends on the data pattern. If we have a data pattern of all 1s or all 0s, the signal rate may be different from a data pattern of alternating 0s and 1s.

The relationship between data rate (*N*) and signal rate (*S*), we need to define three cases: the worst, best, and average.

- The worst case is when we need the maximum signal rate.
- The best case is when we need the minimum.
- In data communications, we are usually interested in the average case.

$$S_{ave} = c \times N \times (1/r) \quad \text{baud}$$

N is the data rate (bps);
c is the case factor, which varies for each case (worst, best & avg.)
S is the number of signal elements per second;
r is the previously defined factor.

Example 4.1

A signal is carrying data in which one data element is encoded as one signal element (r = 1/1). If the bit rate is 100 kbps, what is the average value of the baud rate if c is between 0 and 1?

Solution

We assume that the average value of c is 1/2 . The baud rate is then

$$\boxed{S = c \times N \times \frac{1}{r} = \frac{1}{2} \times 100{,}000 \times \frac{1}{1} = 50{,}000 = 50 \text{ kbaud}}$$

Characteristics of line coding schemes

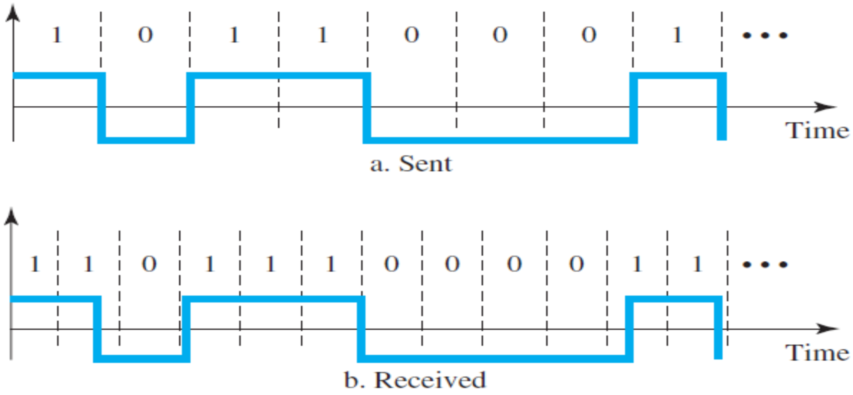
- ***Self-synchronization***

- The clocks at the sender and the receiver must have the same bit interval.

- If the receiver clock is faster or slower it will misinterpret the incoming bit stream.

- The sender sends 10110001, while the receiver receives 110111000011.

Figure 4.3 *Effect of lack of synchronization*



A self-synchronizing digital signal includes ***timing information*** **in the data being transmitted**. This can be achieved if there are transitions in **the signal that alert the receiver to the beginning, middle, or end of the pulse**. If the receiver's clock is out of synchronization, these points can reset the clock.

Characteristics of line coding schemes

Self-synchronization

In a digital transmission, the receiver clock is 0.1 percent faster than the sender clock. How many extra bits per second does the receiver receive if the data rate is 1 kbps?

1. 1 bps
2.10 bps
3.100 bps
4.1000 bps

At 1 kbps bits send by sender = 1000

Receiver clock is 0.1% faster than sender clock. so Extra bits received by receiver = 1000 * 0.1/100 =1 bit

Total bits received by received= 1001 bits

Extra bit per second received by receiver= 1 bps

Characteristics of line coding schemes

Self-synchronization

In a digital transmission, the receiver clock is 0.1 percent faster than the sender clock. How many extra bits per second does the receiver receive if the data rate is 1 Mbps?

1. 10 bps
2. 100 bps
3. 1000 bps
4. 10000 bps

Characteristics of line coding schemes

Self-synchronization

In a digital transmission, the receiver clock is 0.1 percent faster than the sender clock. How many extra bits per second does the receiver receive if the data rate is 1 Mbps?

1. 10 bps
2. 100 bps
3. **1000 bps**
4. 10000 bps

At 1 Mbps bits send by sender = 1000000

Receiver clock is 0.1% faster than sender clock. so Extra bits received by receiver = 1000000 * 0.1/100 =1000 bits

Total bits received by received= 1001000 bits

Extra bit per second received by receiver= 1000 bps

Characteristics of line coding schemes

- ***Baseline Wandering***
  - In decoding a digital signal, the receiver calculates a running average of the received signal power. This average is called the baseline.

    Average signal power ----> Baseline

  - The incoming signal power is evaluated against this baseline to determine the value of the data element.
  - A long string of 0s or 1s can cause a drift in the baseline **(baseline wandering)** and make it difficult for the receiver to decode correctly. So, we need to select a line coding scheme that eliminates this problem.

In decoding a digital signal, the receiver calculates a running average of the received signal power, called the ________.

a. baseline

b. base

c. line

d. broadline

Characteristics of line coding schemes

- ***DC Components***

When the voltage level in a digital signal is constant for a while, the spectrum creates very low frequencies. These frequencies around zero, called DC (direct-current) components, present problems for a system that cannot pass low frequencies or a system that uses electrical coupling (via a transformer). For these systems, we need a scheme with no DC component. The signal with a DC component cannot pass through a transformer.

E.g., in a binary system that transmits a '1' as +1 volt and a '0' as -1 volt, making the total number of '0' and '1' bits roughly equal over some interval gives a DC balanced signal.

- ***Complexity***

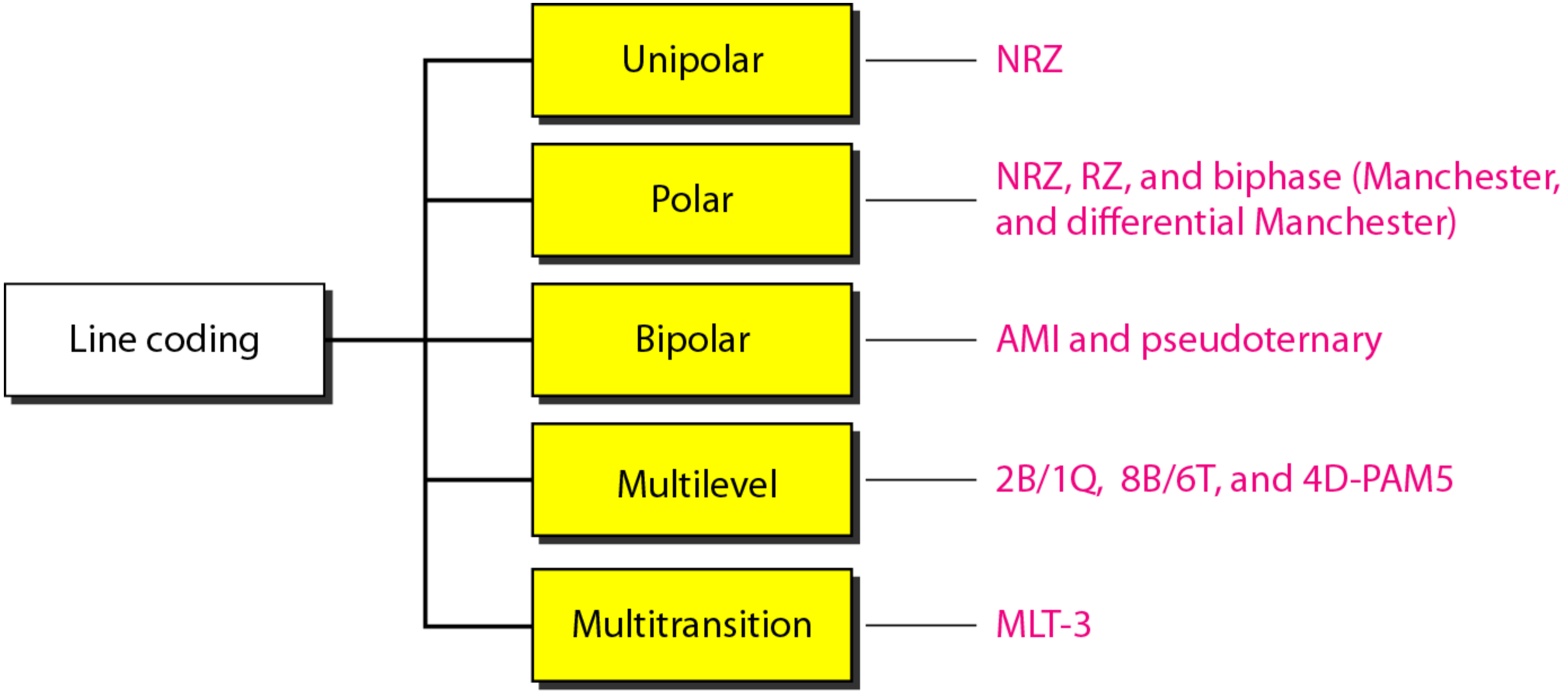
A complex scheme is more costly to implement than a simple one. For example, a scheme that uses four signal levels is more difficult to interpret than one that uses only two levels.

- ***Immunity to Noise and Interference***

Another desirable code characteristic is a code that is immune to noise and other interferences.

Line Coding Schemes

Line coding converts a sequence of digital bits to a digital signal



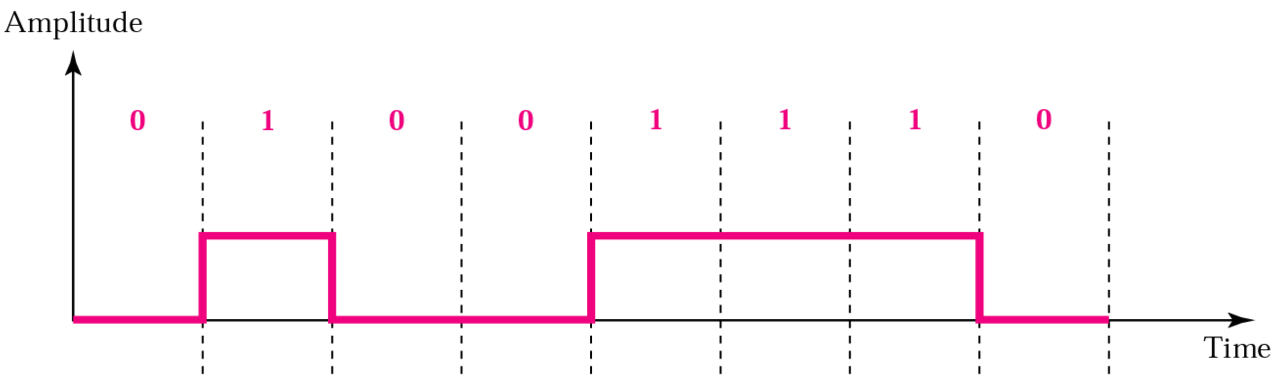
(Digital Signal) Line Coding Schemes

Unipolar Scheme

In a **unipolar** scheme, all the signal levels are on one side of the time axis, either above or below. Unipolar encoding uses only one voltage level.

NRZ (Non-Return-to-Zero)

- The positive voltage defines bit 1 and the zero voltage defines bit 0. The signal level does not return to zero during a symbol transmission.



(Digital Signal) Line Coding Schemes

Polar Schemes

- In **polar** schemes, the voltages are on both sides of the time axis
- The voltage level for 0 can be positive and the voltage level for 1 can be negative.

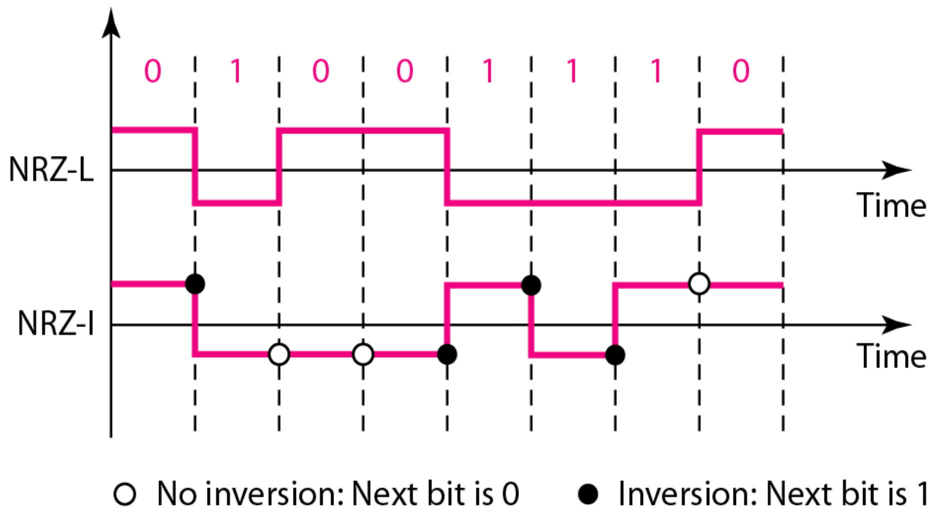
- There are two versions:

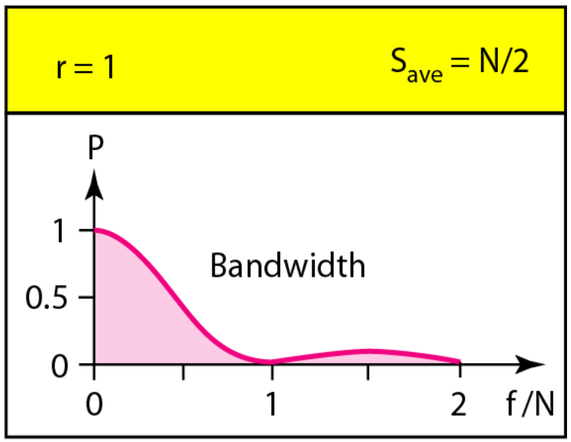
  NZR - Level (NRZ-L) - the level of the voltage determines the value of the bit. positive voltage for one symbol and negative for the other

  +ve volt encodes 0

  –ve volt encodes 1

  NRZ - Inversion (NRZ-I) - the change or lack of change in the level of the voltage determines the value of the bit. If there is no change, the bit is 0; if there is a change, the bit is 1.





Note on Unipolar and Polar Line Coding

Unipolar Non return to zero (NRZ)

- Uses more power
- For continuous set of zeros or ones there will be self-synchronization and base line wandering problem.

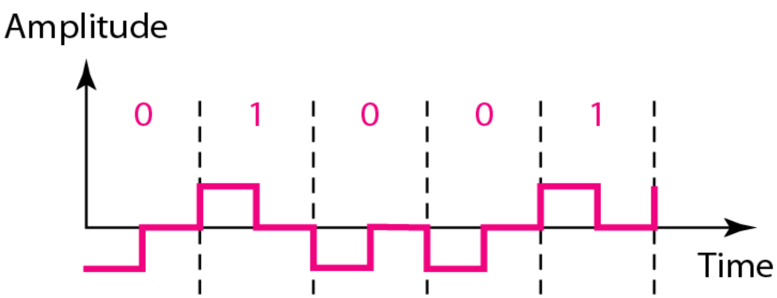
Polar NRZ-L and NRZ-I

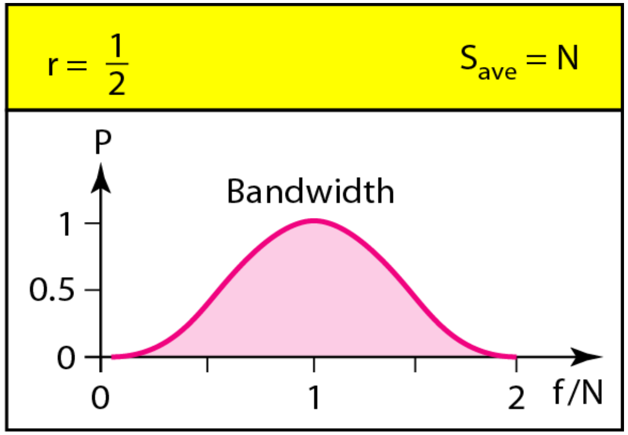
- Both have an average signal rate of N/2 Bd. Where N is the data rate
- DC component problem and baseline wandering
- Both have no self synchronization & no error detection.
- Both are relatively simple to implement.

Return-to-Zero (RZ)

- It uses three values: positive, negative, and zero.

- The signal goes to 0 in the middle of each bit. It remains there until the beginning of the next bit.





Note

- This scheme has more signal transitions (two per symbol) and therefore requires a wider bandwidth.

- No DC components or baseline wandering.

- Self synchronization - transition indicates symbol value.

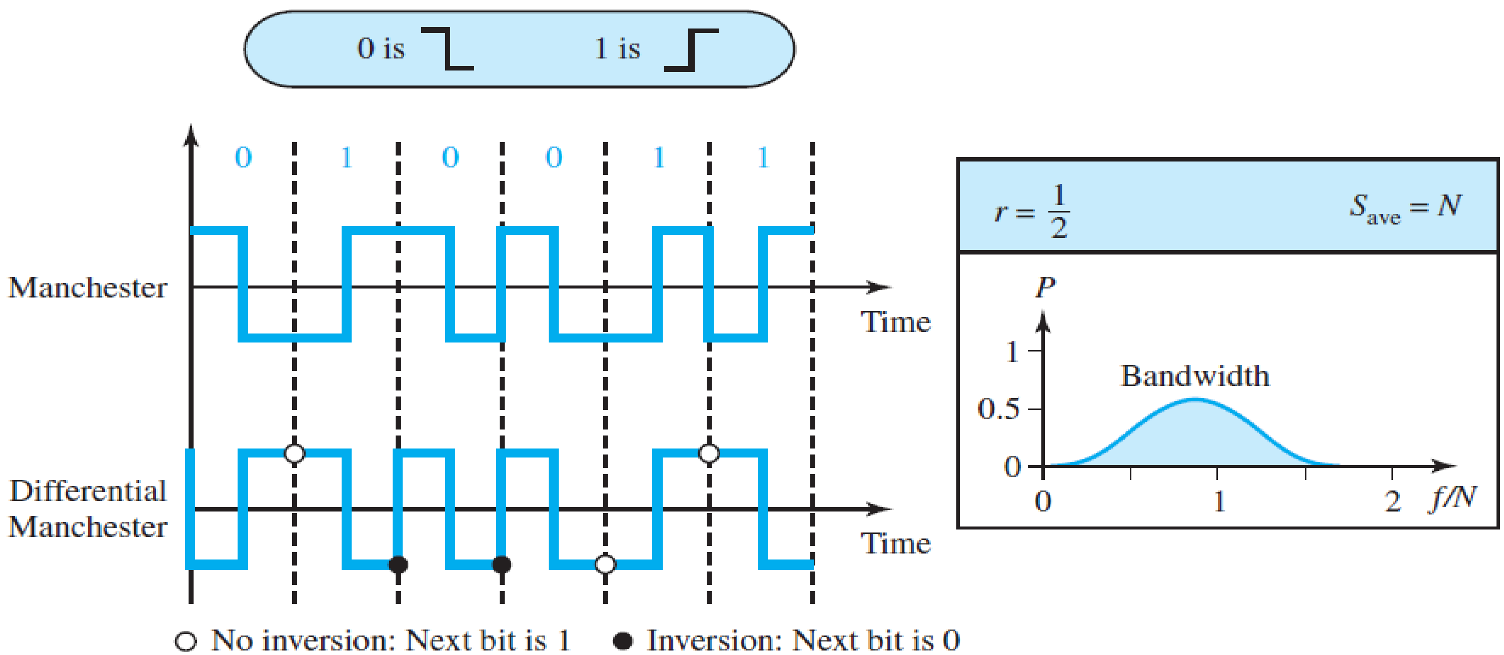
- More complex as it uses three voltage level. It has no error detection capability.

Biphase: Manchester and Differential Manchester

Manchester encoding, the duration of the bit is divided into two halves. The voltage remains at one level during the first half and moves to the other level in the second half.

Differential Manchester, If the next bit is 0, there is a transition; if the next bit is 1, there is none.

Figure 4.8 *Polar biphase: Manchester and differential Manchester schemes*



Bipolar Schemes

- In **bipolar** encoding (sometimes called ***multilevel binary***), there are three voltage levels:positive, negative, and zero.

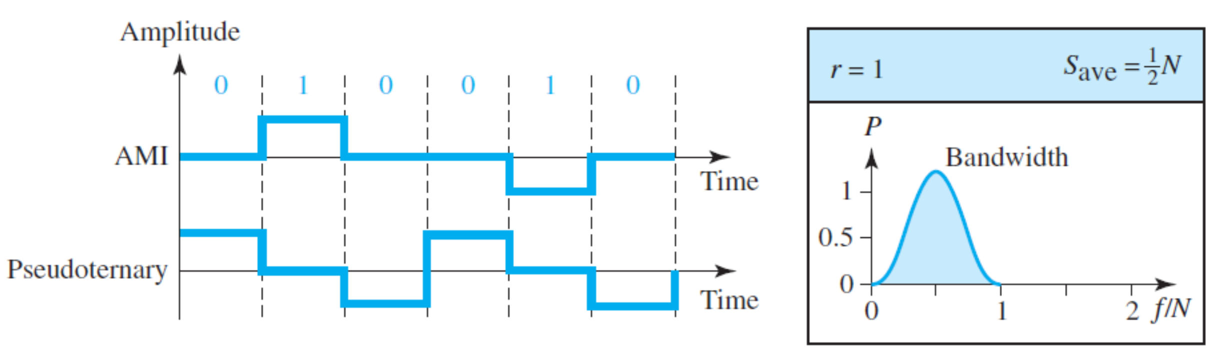
Alternate mark inversion (AMI)

- AMI means alternate 1 inversion. A neutral zero voltage represents binary 0. Binary 1s are represented by alternating positive and negative voltages.

Pseudoternary

- The 1 bit is encoded as a zero voltage and the 0 bit is encoded as alternating positive and negative voltages.

Figure 4.9 *Bipolar schemes: AMI and pseudoternary*



Multilevel Schemes

The code designers have classified these types of coding as *mBnL*,
where *m* is the length of the binary pattern,
B means binary data,
n is the length of the signal pattern, and
L is the number of levels in the signaling.

A letter is often used in place of *L*:
B (binary) for *L* = 2,
T (ternary) for *L* = 3, and
Q (quaternary) for *L* = 4.

2B1Q **(two binary, one quaternary)**

Multilevel Schemes

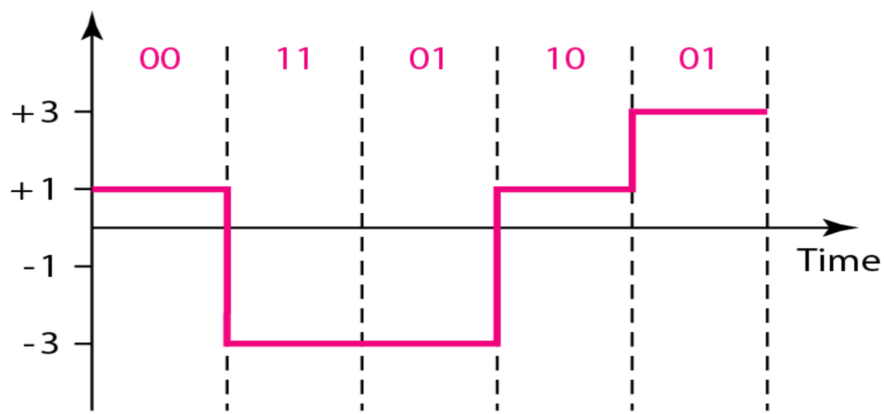
2B1Q **(two binary, one quaternary)**

$m = 2$, $n = 1$, and $L = 4$ (quaternary).

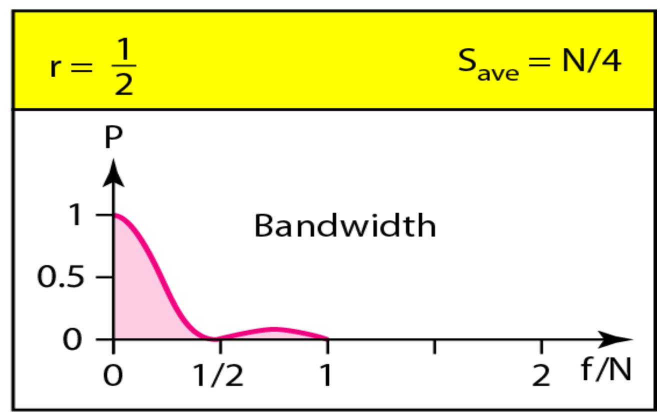
0011011001

	Previous level: positive	Previous level: negative
Next bits	Next level	Next level
00	+1	-1
01	+3	-3
10	-1	+1
11	-3	+3

Transition table



Assuming positive original level



Note: The 2B1Q scheme is used in DSL (Digital Subscriber Line) technology to provide a high-speed connection to the Internet by using subscriber telephone lines

Multilevel Schemes

2B1Q

Digitaldata: 11001100101110

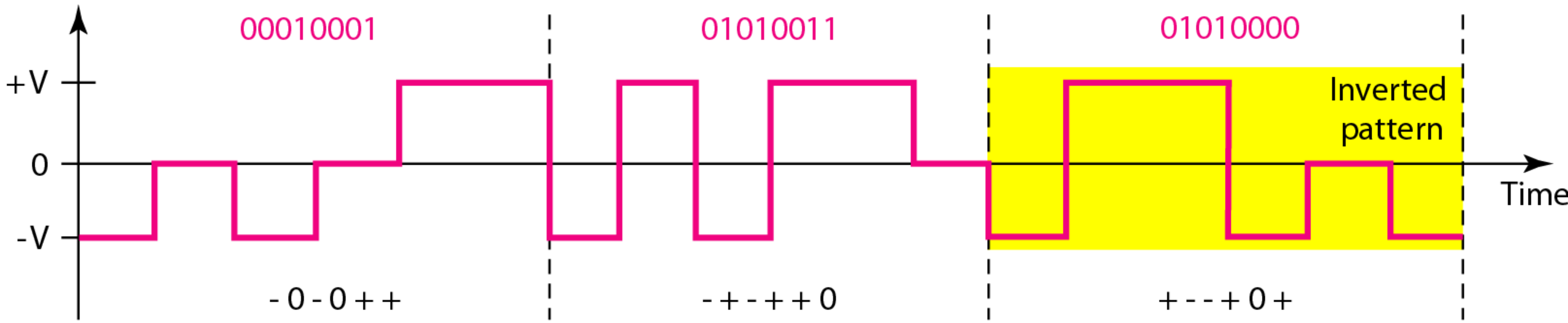
	Previous level: positive	Previous level: negative
Next bits	**Next level**	**Next level**
00	+1	-1
01	+3	-3
10	-1	+1
11	-3	+3

Transition table

Multilevel Schemes

8B6T (eight binary, six ternary)

- This code is used with 100BASE-4T cable
- The idea is to encode a pattern of 8 bits as a pattern of six signal elements, where the signal has three levels (ternary).
- 2^8 = 256 different data patterns and 3^6 = 729 different signal patterns.



Three data patterns encoded as three signal patterns

Each signal pattern has a weight of 0 or +1 DC values. This means that there is no pattern with the weight −1. To make the whole stream DC-balanced, the sender keeps track of the weight. If two groups of weight 1 are encountered one after another, the first one is sent as is, while the next one is totally inverted to give a weight of −1.

APPENDIX F

8B/6T Code

This appendix is a tabulation of 8B/6T code pairs. The 8-bit data are shown in hexadecimal format. The 6T code is shown as + (positive signal), – (negative signal), and 0 (lack of signal) notation.

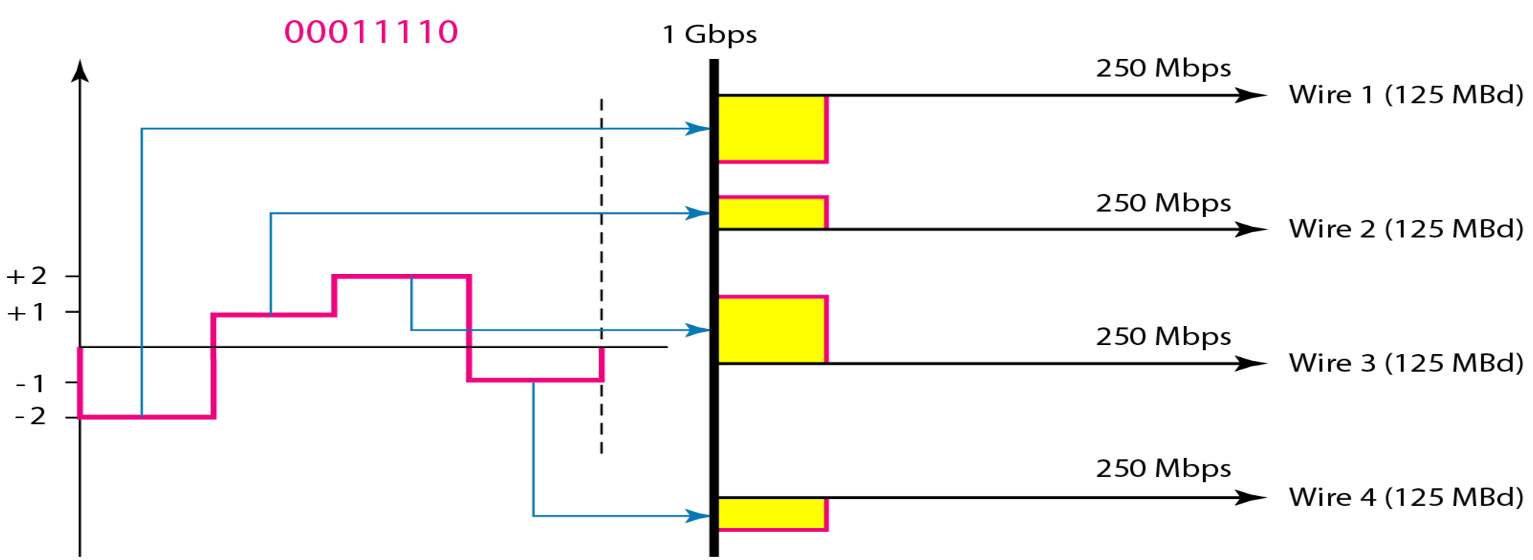
Table F.1 *8B/6T code*

Data	*Code*	*Data*	*Code*	*Data*	*Code*	*Data*	*Code*
00	−+00−+	**20**	−++−00	**40**	−00+0+	**60**	0++0−0
01	0−+−+0	**21**	+00+−−	**41**	0−00++	**61**	+0+−00
02	0−+0−+	**22**	−+0−++	**42**	0−0+0+	**62**	+0+0−0
03	0−++0−	**23**	+−0−++	**43**	0−0++0	**63**	+0+00−
04	−+0+0−	**24**	+−0+00	**44**	−00++0	**64**	0++00−
05	+0−−+0	**25**	−+0+00	**45**	00−0++	**65**	++0−00
06	+0−0−+	**26**	+00−00	**46**	00−+0+	**66**	++00−0
07	+0−+0−	**27**	−+++−−	**47**	00−++0	**67**	++000−
08	−+00+−	**28**	0++−0−	**48**	00+000	**68**	0++−+−
09	0−++−0	**29**	+0+0−−	**49**	++−000	**69**	+0++−−
0A	0−+0+−	**2A**	+0+−0−	**4A**	+−+000	**6A**	+0+−+−
0B	0−+−0+	**2B**	+0+−−0	**4B**	−++000	**6B**	+0+−−+
0C	−+0−0+	**2C**	0++−−0	**4C**	0+−000	**6C**	0++−−+
0D	+0−+−0	**2D**	++00−−	**4D**	+0−000	**6D**	++0+−−
0E	+0−0+−	**2E**	++0−0−	**4E**	0−+000	**6E**	++0−+−
0F	+0−−0+	**2F**	++0−−−0	**4F**	−0+000	**6F**	++0−−+
10	0−−+0+	**30**	+−00−+	**50**	+−−+0+	**70**	000++−
11	−0−0++	**31**	0+−−+0	**51**	−+−0++	**71**	000+−+
12	−0−+0+	**32**	0+−0−+	**52**	−+−+0+	**72**	000−++
13	−0−++0	**33**	0+−+0−	**53**	−+−++0	**73**	000+00
14	0−−++0	**34**	+−0+0−	**54**	+−−++0	**74**	000+0−
15	−−00++	**35**	−0+−+0	**55**	−−+0++	**75**	000+−0
16	−−0+0+	**36**	−0+0−+	**56**	−−++0+	**76**	000−0+
17	−−0++0	**37**	−0++0−	**57**	−−++0	**77**	000−+0
18	−+0−+0	**38**	+−00+−	**58**	−+−0++	**78**	+++−−0

Multilevel Schemes

4D-PAM5 four-dimensional five level pulse amplitude modulation

- 4D means that data is sent over four wires at the same time
- It uses five voltage levels, such as −2, −1, 0, 1, and 2
- All 8 bits can be fed into a wire simultaneously and sent by using one signal element. The point here is that the four signal elements comprising one signal group are sent simultaneously in **a four dimensional setting**.
- Gigabit LANs use this encoding technique to transmit 1 Gbps data over four copper cables which can handle 125 Mbaud.



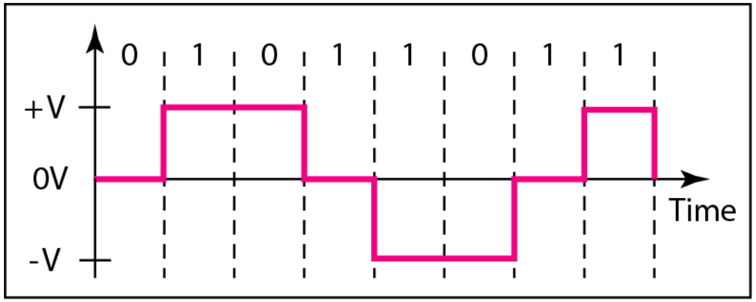
Multitransition: MLT-3

The **multiline transmission, three-level (MLT-3) scheme uses** three levels (+*V*, 0, and −*V*) and three transition rules to move between the levels.

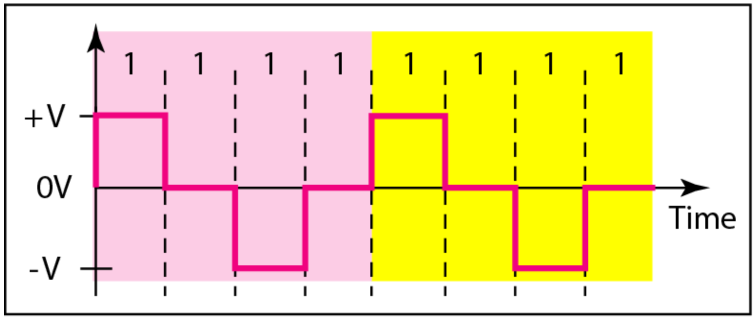
1. If the next bit is 0, there is no transition.

2. If the next bit is 1 and the current level is not 0, the next level is 0.

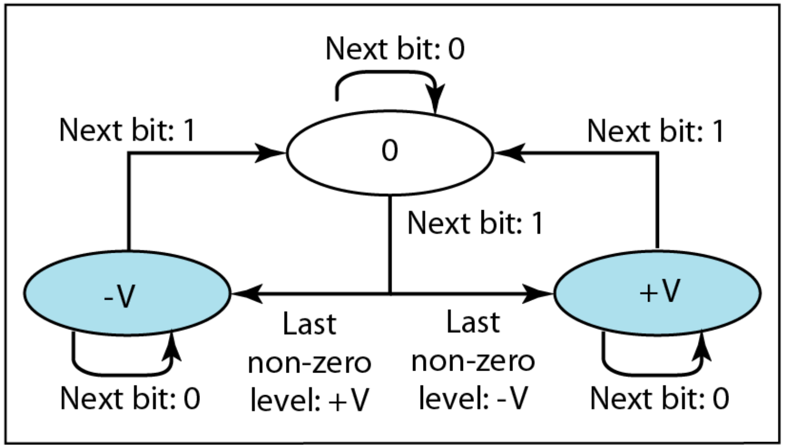
3. If the next bit is 1 and the current level is 0, the next level is the opposite of the last nonzero level.



a. Typical case



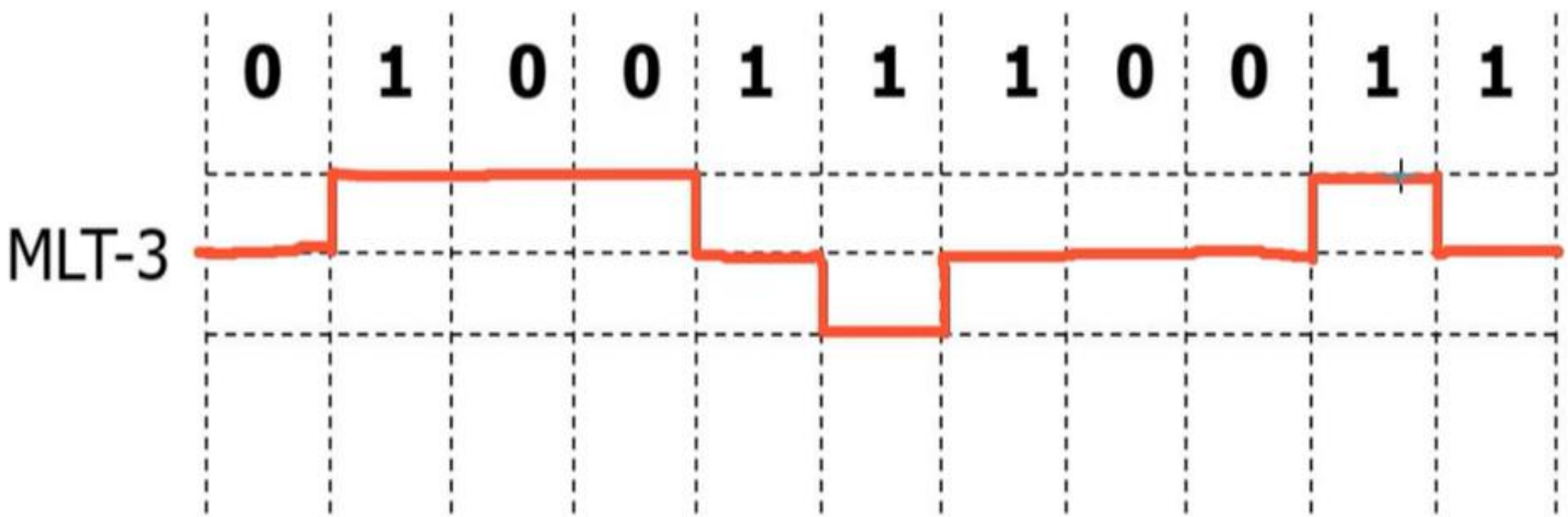
b. Worse case



c. Transition states

Multitransition: MLT-3

1. If the next bit is 0, there is no transition.
2. If the next bit is 1 and the current level is not 0, the next level is 0.
3. If the next bit is 1 and the current level is 0, the next level is the opposite of the last nonzero level.



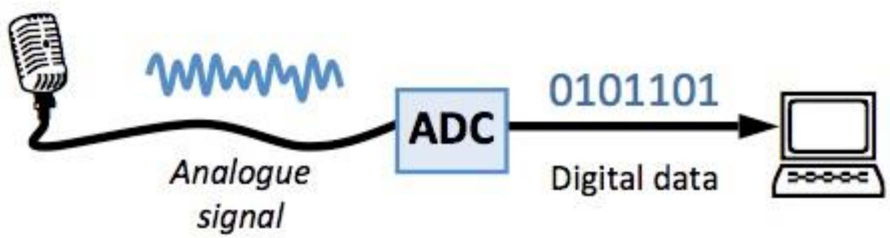
+ve neutral -ve

Consider the binary sequence: 011101110.
Draw the line coding graph for the following:
NRZ-L
NRZ-I
Polar RZ
Manchester encoding
Differential Manchester encoding
AMI encoding
Psuedometry encoding

Table 4.1 *Summary of line coding schemes*

Category	*Scheme*	*Bandwidth (average)*	*Characteristics*
Unipolar	NRZ	$B = N/2$	Costly, no self-synchronization if long 0s or 1s, DC
Polar	NRZ-L	$B = N/2$	No self-synchronization if long 0s or 1s, DC
	NRZ-I	$B = N/2$	No self-synchronization for long 0s, DC
	Biphase	$B = N$	Self-synchronization, no DC, high bandwidth
Bipolar	AMI	$B = N/2$	No self-synchronization for long 0s, DC
Multilevel	2B1Q	$B = N/4$	No self-synchronization for long same double bits
	8B6T	$B = 3N/4$	Self-synchronization, no DC
	4D-PAM5	$B = N/8$	Self-synchronization, no DC
Multitransition	MLT-3	$B = N/3$	No self-synchronization for long 0s

ANALOG-TO-DIGITAL CONVERSION

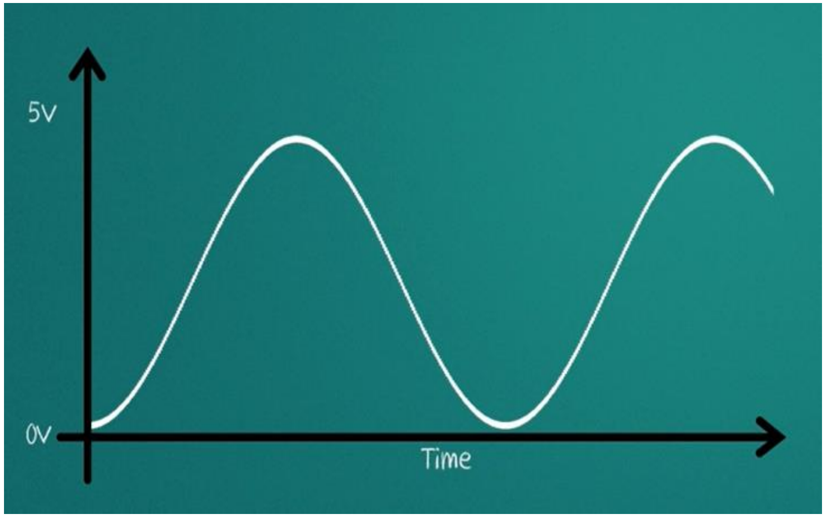


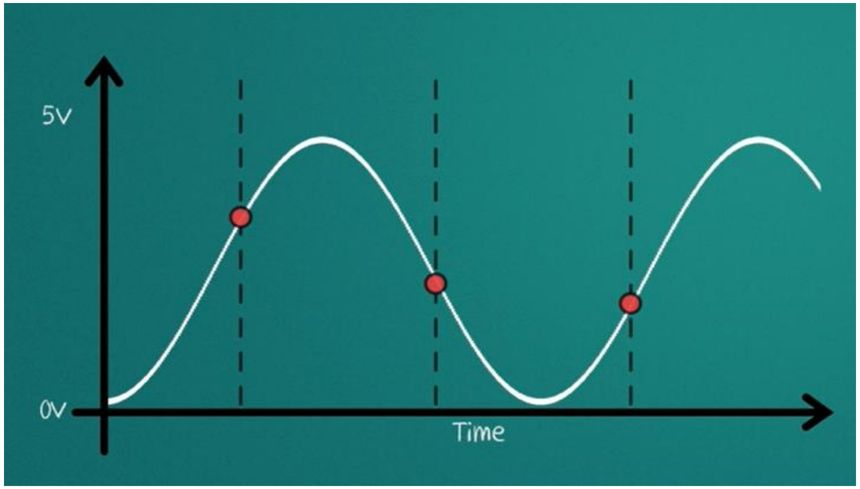
A digital signal is superior to an analog signal because it is more robust to noise and can easily be recovered, corrected and amplified. For this reason, the tendency today is to change an analog signal to digital data

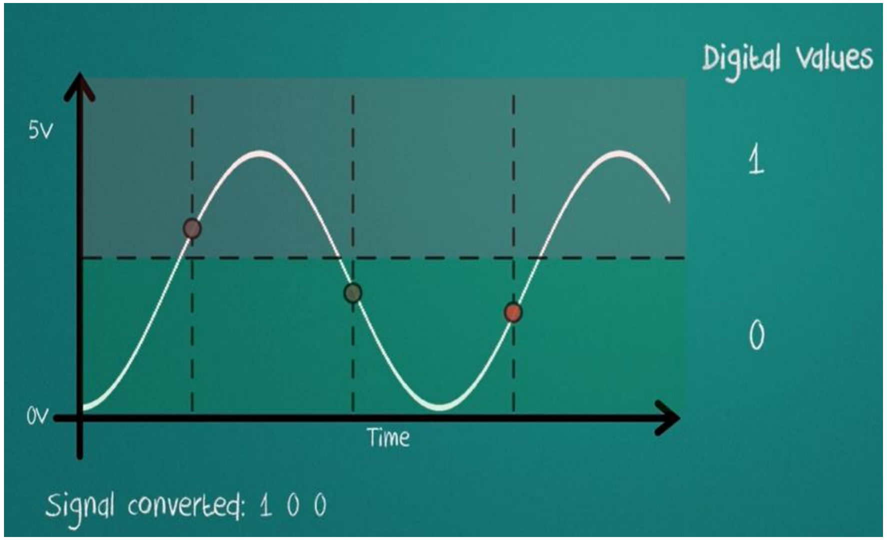
Techniques

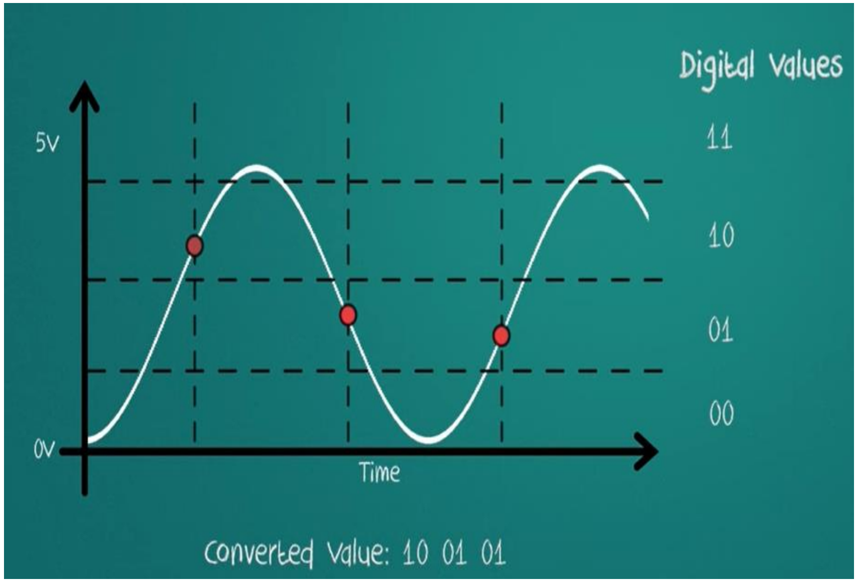
1. Pulse Code Modulation (PCM)
2. Delta Modulation (DM)

Example


5V
0V
Time

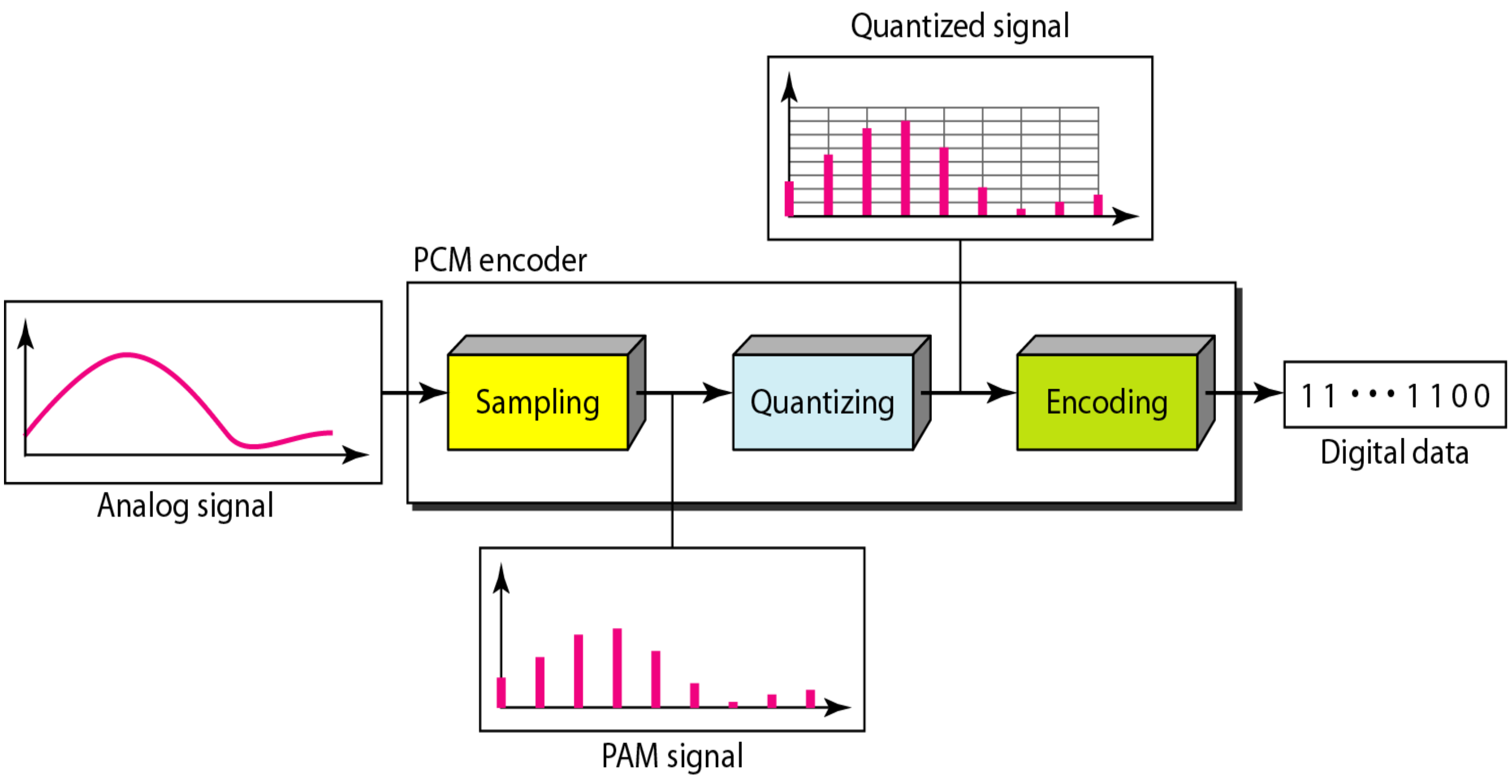

5V
0V
Time


Digital Values
5V
1
0
0V
Time
Signal converted: 1 0 0


Digital Values
5V
11
10
01
00
0V
Time
Converted Value: 10 01 01

Pulse Code Modulation (PCM)

- A technique by which analog signal gets converted into digital form
- A PCM encoder has three processes:
  1. Sampling
  2. Quantization
  3. Binary encoding



Sampling

- The process of obtaining amplitudes of a signal at regular intervals.
- Analog signal is sampled every T_S secs.
- T_s is referred to as the sampling interval.
- $f_s = 1/T_s$ is called the sampling rate or sampling frequency (is the number of samples per second).
- There are 3 sampling methods:
  - Ideal - an impulse at each sampling instant
  - Natural - a pulse of short width with varying amplitude
  - Flattop - sample and hold, like natural but with single amplitude value
- The sampling process is sometimes referred to as *pulse amplitude modulation (PAM)*..

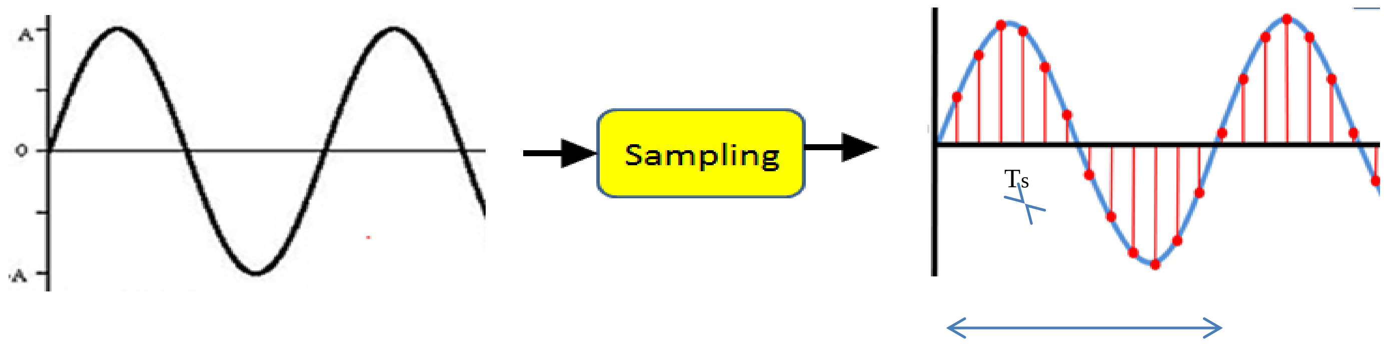
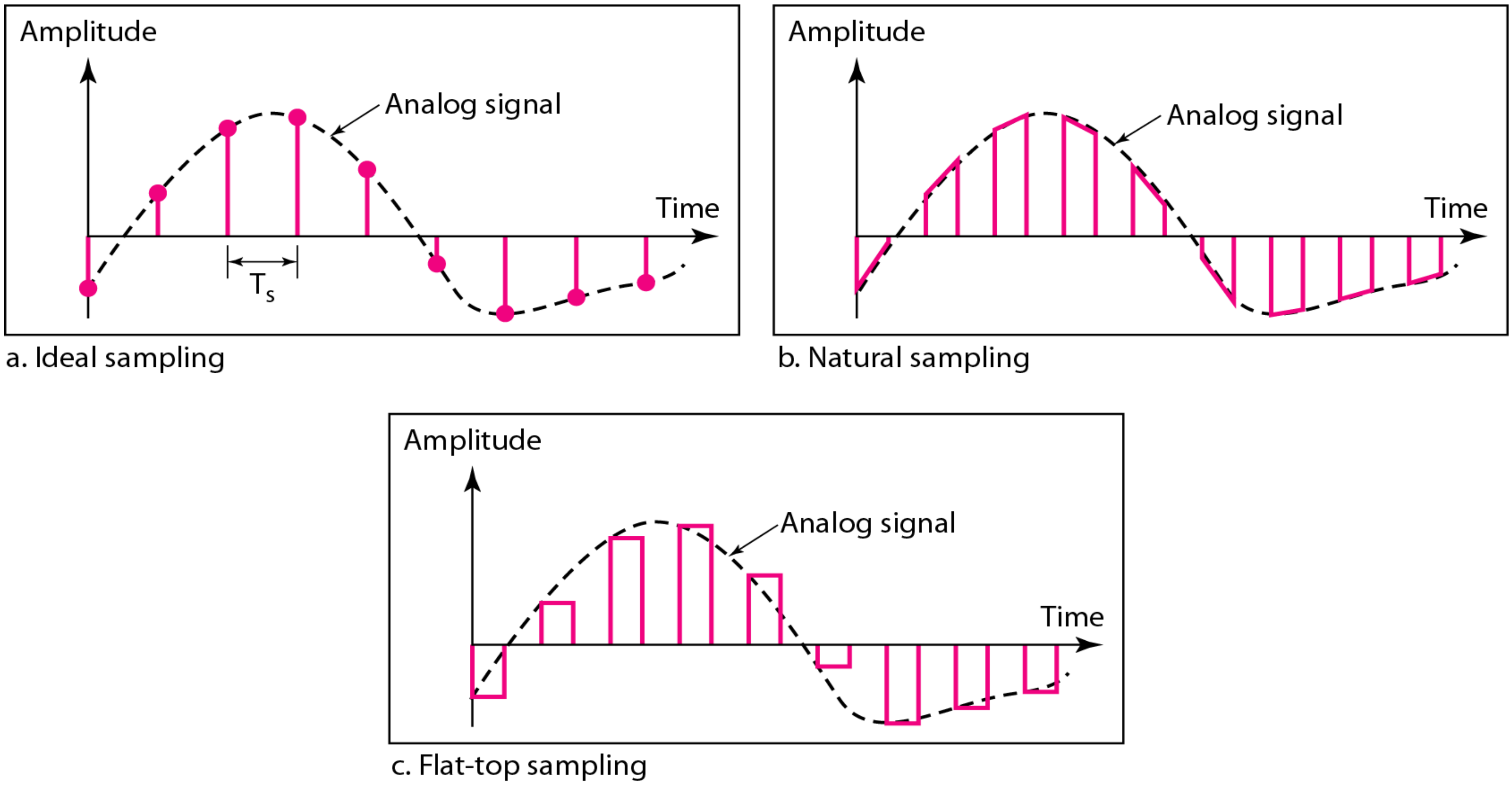
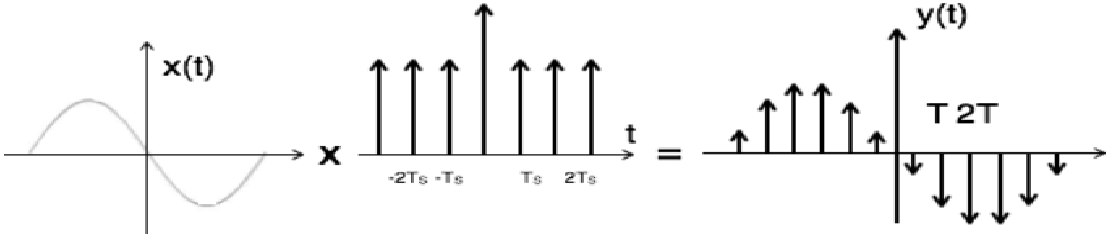


Figure 4.22 *Three different sampling methods for PCM*

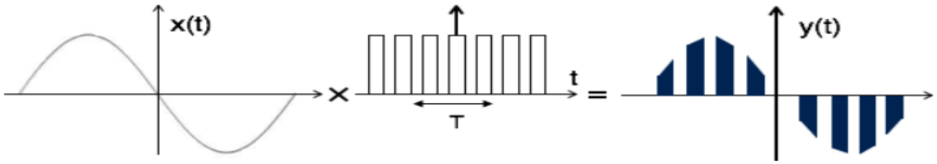


Note

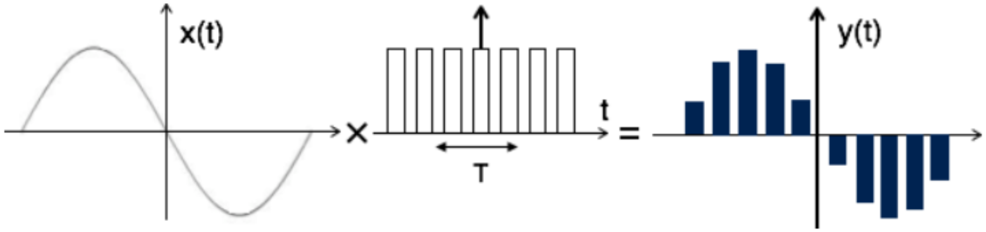
Impulse/Ideal sampling can be performed by multiplying input signal x(t) with impulse train of period 'T'. Here, the amplitude of impulse changes with respect to amplitude of input signal x(t)



Natural sampling is similar to impulse sampling, except the impulse train is replaced by pulse train of period T.



Flat top: Here, the top of the samples are flat i.e. they have constant amplitude. Hence, it is called as flat top sampling or practical sampling. Flat top sampling makes use of sample and hold circuit.



Sampling Rate

Note

According to the Nyquist theorem, the sampling rate must be at least 2 times the highest frequency contained in the signal.

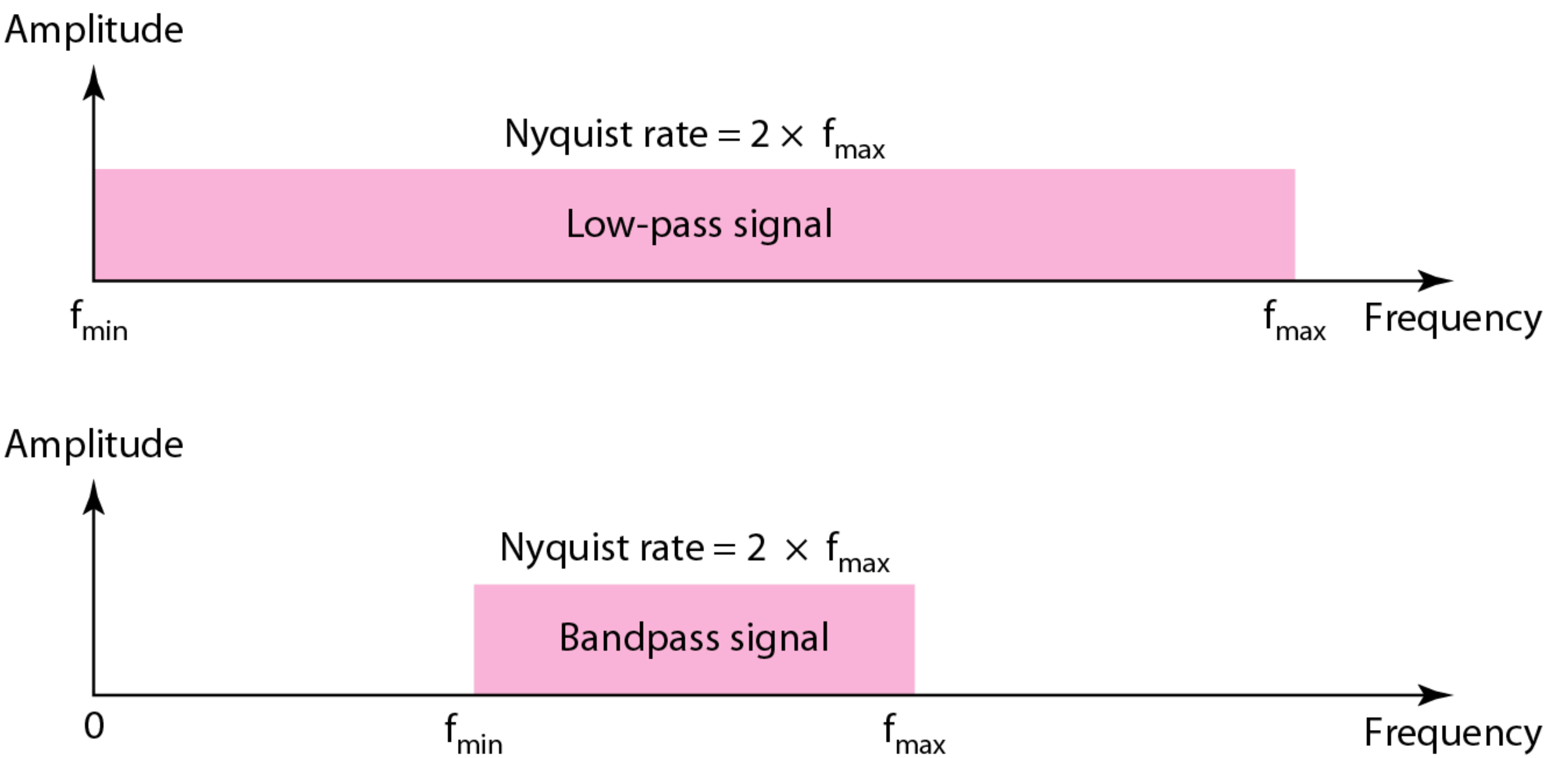
Why so?

If the sampling is performed at a proper rate, no info is lost about the original signal and it can be properly reconstructed later on.

$$f_s \geq 2 f_{\max}$$

Sampling rate for telephone : 8Khz (8000 samples per second)
Sampling rate for Mp3 : 44Khz
Sampling rate for Blueray disc :1Mhz

Figure 4.23 *Nyquist sampling rate for low-pass and bandpass signals*



Example 4.6

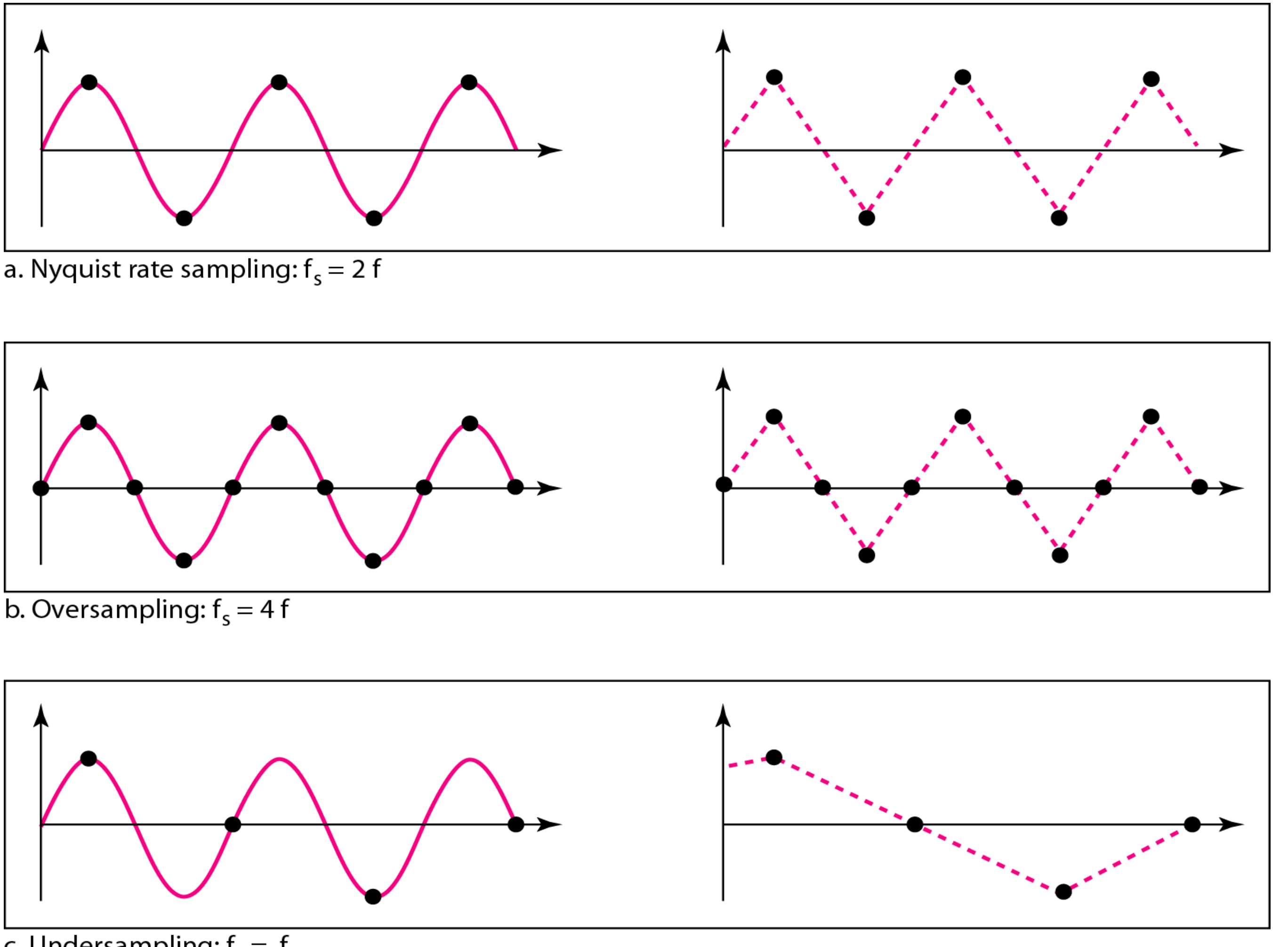
For an intuitive example of the Nyquist theorem, let us sample a simple sine wave at three sampling rates:
$f_s = 4f$ (2 times the Nyquist rate),
$f_s = 2f$ (Nyquist rate), and
$f_s = f$ (one-half the Nyquist rate). Figure 4.24 shows the sampling and the subsequent recovery of the signal.

It can be seen **that sampling at the Nyquist rate can create a good approximation of the original sine wave** (part a).

Oversampling in part b can also create the same approximation, but it is redundant and unnecessary.

Sampling below the Nyquist rate (part c) does not produce a signal that looks like the original sine wave.

Figure 4.24 *Recovery of a sampled sine wave for different sampling rates*



a. Nyquist rate sampling: $f_s = 2\,f$

b. Oversampling: $f_s = 4\,f$

c. Undersampling: $f_s = f$

Example 4.9

Telephone companies digitize voice by assuming a maximum frequency of 4000 Hz. The sampling rate therefore is 2x4000= 8000 samples per second.

Example 4.10

A complex low-pass signal has a bandwidth of 200 kHz. What is the minimum sampling rate for this signal?

Solution

The bandwidth of a low-pass signal is between 0 and f, where f is the maximum frequency in the signal. Therefore, we can sample this signal at 2 times the highest frequency (200 kHz). The sampling rate is therefore 400,000 samples per second.

Example 4.11

A complex bandpass signal has a bandwidth of 200 kHz. What is the minimum sampling rate for this signal?

Solution

We cannot find the minimum sampling rate in this case because we do not know where the bandwidth starts or ends. We do not know the maximum frequency in the signal.

Example 4.9

Let us assume, VOIP application digitize voice by assuming a maximum frequency of 7000 Hz. What is the sampling rate?

Ans:2*7000=14000 samples per second

What is the Nyquist sampling rate for each of the following signals?

i) A low pass signal with bandwidth of 150 kHz.

ii) A band pass signal with bandwidth of 200 kHz having lowest frequency of 100 kHz.

What is the Nyquist sampling rate for each of the following signals?
i) A low pass signal with bandwidth of 150 kHz.
ii) A band pass signal with bandwidth of 200 kHz having lowest frequency of 100 kHz.

i) Low-pass signal f_l=0 Hz $f_h = f_l$ + BW = 0+ 150kHz = 150KHz
Nyquist Sampling rate = 2 x 150k= 300, 000 sample per second

ii) Band-pass signal $f_h = f_l$ + BW = 100K + 200K = 300kHz
Nyquist Sampling rate = 2 x 300k = 600, 000 sample per second

Quantization

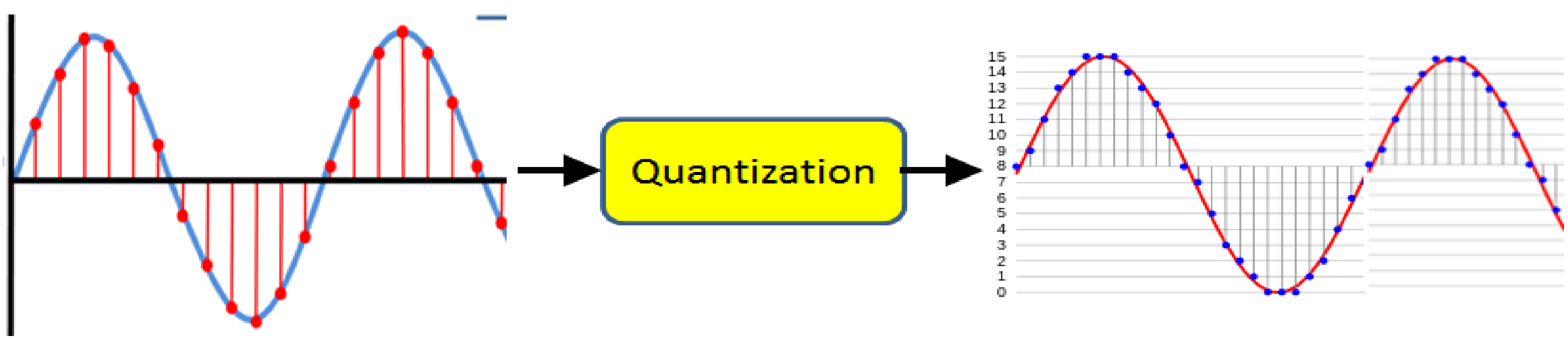
- Sampling results in a series of pulses of varying amplitude values ranging between two limits: a min and a max.
- The amplitude values are infinite between the two limits.
- We need to map the *infinite* amplitude values onto a finite set of known values. So, we have to make discrete amplitude levels and sample at any point can take only these values (Rounding off of the signal to its nearest level-quantization level)

The following are the steps in quantization:

1. We assume that the original analog signal has instantaneous amplitudes between V_{min} and V_{max}.
2. We divide the range into L zones, each of height Δ (delta).

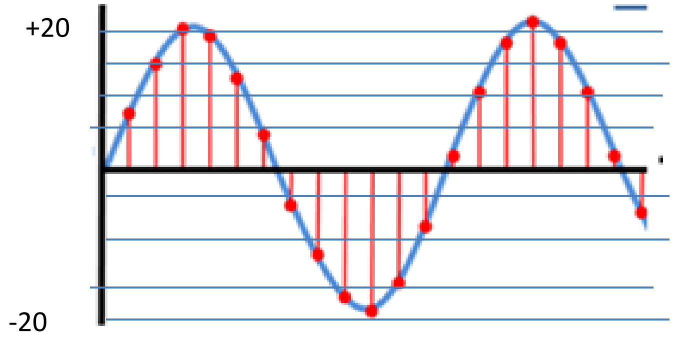
$$\Delta = \frac{V_{\max} - V_{\min}}{L}$$

3. We assign quantized values of 0 to $L - 1$ to the midpoint of each zone.
4. We approximate the value of the sample amplitude to the quantized values.



Quantization

- Assume we have a voltage signal with amplitudes V_{min}=-20V and V_{max}=+20V.
- We want to use L=8 quantization levels.
- Zone width $\Delta = ((+20) - (-20))/8 = 5$
- The 8 zones are: -20 to -15, -15 to -10, -10 to -5, -5 to 0, 0 to +5, +5 to +10, +10 to +15, +15 to +20
- The midpoints are: -17.5, -12.5, -7.5, -2.5, 2.5, 7.5, 12.5, 17.5



Quantization

- Assume we have a voltage signal with amplitudes V_{min}=-20V and V_{max}=+20V.

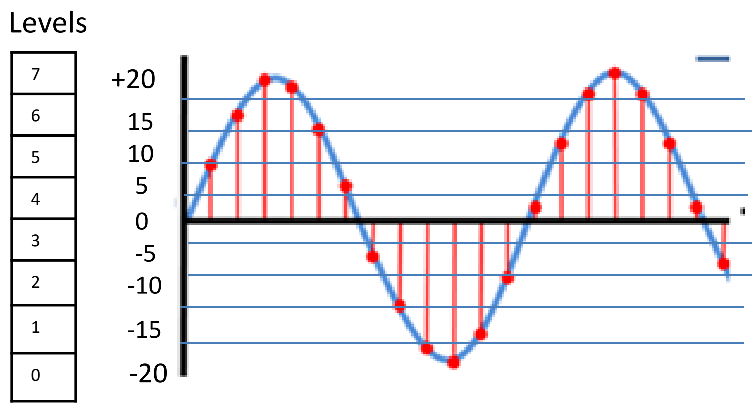
- We want to use L=8 quantization levels.

- Zone width $\Delta = ((+20) - (-20))/8 = 5$

- The 8 zones are: -20 to -15, -15 to -10, -10 to -5, -5 to 0, 0 to +5, +5 to +10, +10 to +15, +15 to +20

- The midpoints are: -17.5, -12.5, -7.5, -2.5, 2.5, 7.5, 12.5, 17.5

Rounding off of the signal to its nearest level-quantization level



Quantization
Assigning Codes to Zones

Encoding

- Each zone is then assigned a binary code.
- The number of bits required to encode the zones, or the number of bits per sample as it is commonly referred to, is obtained as follows:

$$n_b = \log_2 L$$

- Given our example, $n_b = \log_2 8 = 3$
- Digital representation: The 8 zone (or level) codes are therefore: 000, 001, 010, 011, 100, 101, 110, and 111
- Assigning codes to zones:
  - 000 will refer to zone -20 to -15
  - 001 to zone -15 to -10, etc.

101 110

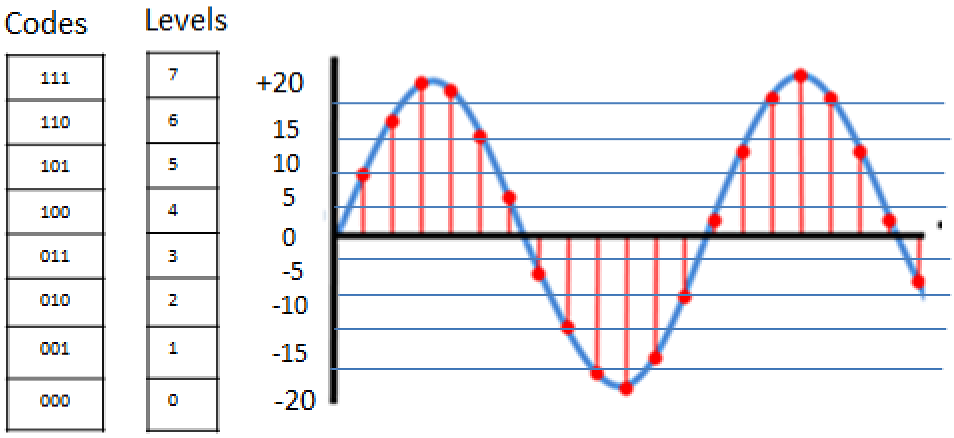
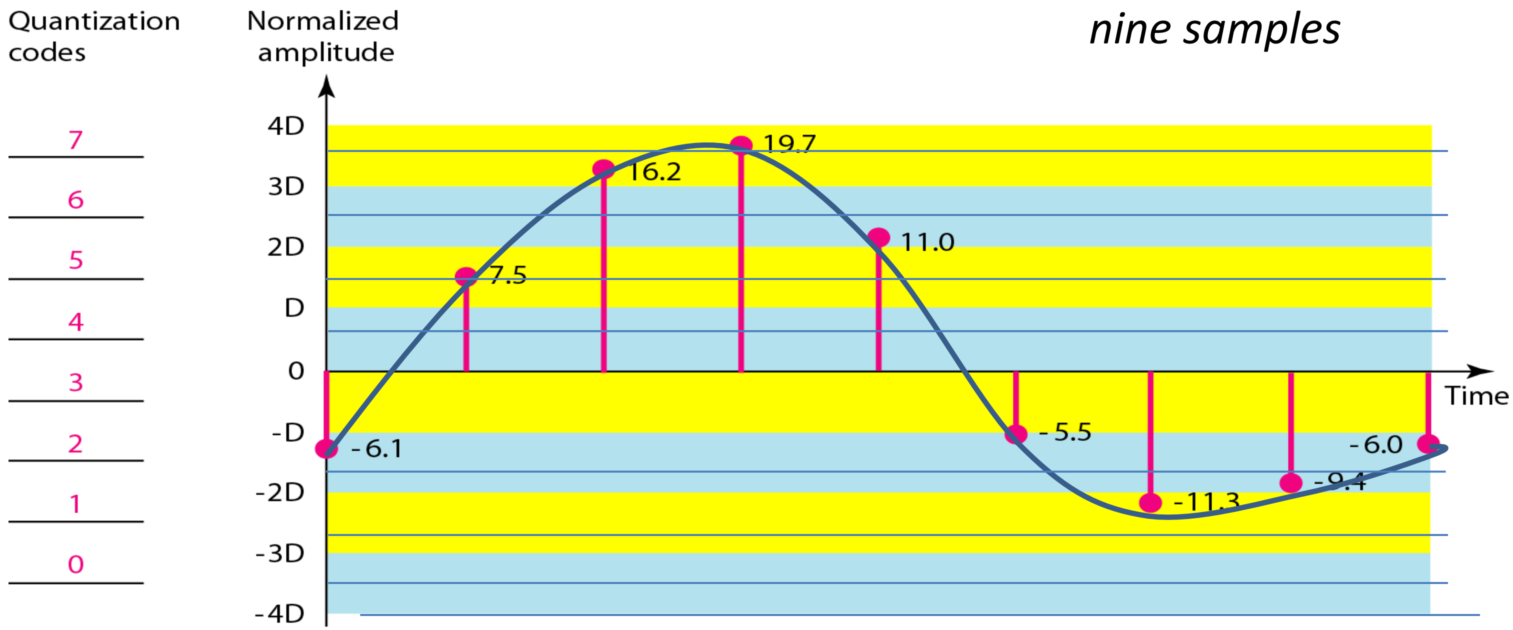


Figure 4.26

Quantization

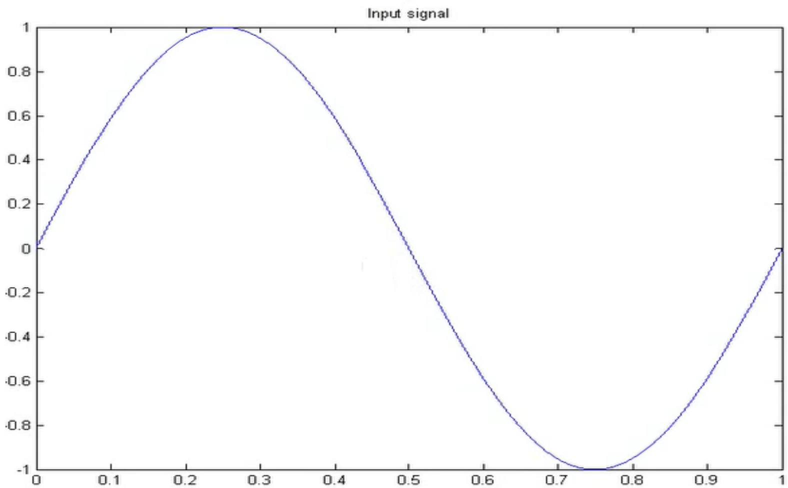


Normalized PAM values	-1.22	1.50	3.24	3.94	2.20	-1.10	-2.26	-1.88	-1.20
Normalized quantized values	-1.50	1.50	3.50	3.50	2.50	-1.50	-2.50	-1.50	-1.50
Normalized error	-0.38	0	+0.26	-0.44	+0.30	-0.40	-0.24	+0.38	-0.30
Quantization code	2	5	7	7	6	2	1	2	2
Encoded words	010	101	111	111	110	010	001	010	010

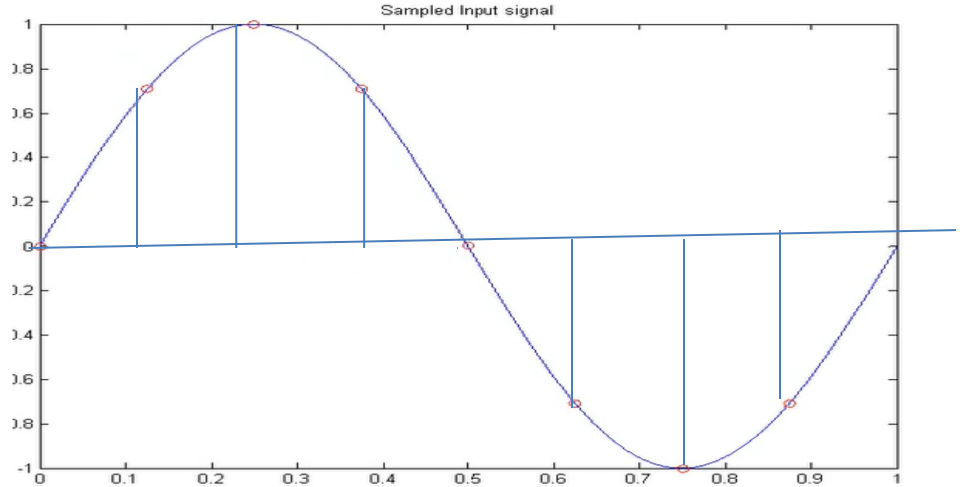
- Quantization is an approximation process.
- The assignment of a specific range of values to signal amplitudes

Note

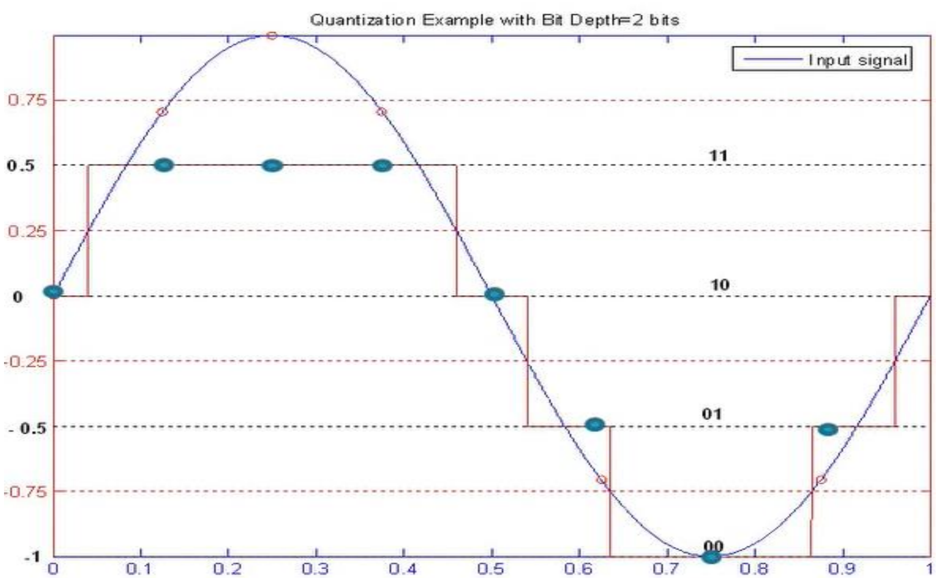
Input Signal



Sampled Input Signal



Sample frequency = 8 samples per second



Quantized using 2 bits = 4 levels (-1, -0.5, 0, +0.5) –quantized values

$$\Delta = ((+1) - (-1))/4 = 0.5$$

$$\log_2 2^2 = 2$$

Sampled Value xq	0	0.707	1	0.707	0	-0.707	-1	-0.707
Quantized Value x	0	0.5	0.5	0.5	0	-0.5	-1	-0.5
Bianry Value	10	11	11	11	00	01	01	01
Quantization Error	0	0.207	0.5	0.207	0	-0.207	0	-0.207

$$e_q = X_q - X$$

10 11 11 11 10 01 00 01

Quantization

Note: The choice of *L*, the number of levels, depends on the range of the amplitudes of the analog signal and how accurately we need to recover the signal.

- If the amplitude of a signal fluctuates between two values only, we need only two levels.
- In audio digitizing, *L* is normally chosen to be 256;
- If the signal, like voice, has many amplitude values, we need more quantization levels.
- In video it is normally thousands.

Choosing lower values of L increases the quantization error if there is a lot of fluctuation in the signal.

Quantization Error

- When a signal is quantized, we introduce an error - the coded signal is an approximation of the actual amplitude value.
- **The difference between actual and coded value (midpoint) is referred to as the quantization error.**
- This can be minimized by increasing the number of quantization levels – more zones.
- The more zones, the smaller Δ which results in smaller errors.
- BUT, the more zones the more bits required to encode the samples -> higher bit rate

Example

- If the input value is also at the middle of the zone, there is no quantization error; otherwise, there is an error.
- The normalized amplitude of the third sample is 3.24, but the normalized quantized value is 3.50. This means that there is an error of 3.50-3.24= +0.26.

Quantization Error and SN_QR

- SNR is actually the ratio of **what is wanted (signal)** to **what is not wanted (noise)**. A high SNR means the signal is less corrupted by noise; a low SNR means the signal is more corrupted by noise.

- Shannon Capacity: To determine the theoretical highest data rate for a noisy channel: $\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$

- The quantization error changes the signal-to-noise ratio of the signal, which in turn reduces the upper limit capacity according to Shannon

- It can be proven that, the Quantization error to the SNRdB of the signal depends on the number of quantization levels *L*, or the bits per sample *nb*,

$$\mathbf{SNR_{dB} = 6.02}\boldsymbol{nb}\mathbf{ + 1.76\ dB}$$

Quantization Error and SN_QR

Example 4.12

- *What is the SNRdB in the example of Figure 4.26?*

We can use the formula to find the quantization. We have eight levels and 3 bits per sample, so SNRdB = 6.02(3) + 1.76 = 19.82 dB.

Increasing the number of levels increases the SNR.

Example 4.13

A telephone subscriber line must have an SNRdB above 40. What is the minimum number of bits per sample?

Solution

We can calculate the number of bits as Telephone companies usually assign 7 or 8 bits per sample.

$$\text{SNRdB} = 6.02n_b + 1.76$$

$$6.02nb+1.76=40$$

$$nb=(40-1.76)/6.02 =6.35$$

Uniform Versus Nonuniform Quantization

S.No	Uniform Quantization	Non-uniform Quantization
1	The quantization step size remains same throughout the dynamic range of the signal	The quantization step size varies with the amplitude of the input signal.
2	SNR ratio varies with the input signal amplitude	SNR ratio can be maintained constant

- In nonuniform, the height of Δ is not fixed; it is greater near the lower amplitudes and less near the higher amplitudes.
- Nonuniform quantization can also be achieved by using a process called **companding and expanding**
- *Companding* means reducing the instantaneous voltage amplitude for large values; expanding is the opposite process.

Companding refers to a technique for compressing and then expanding (or decompressing) an analog or digital signal.

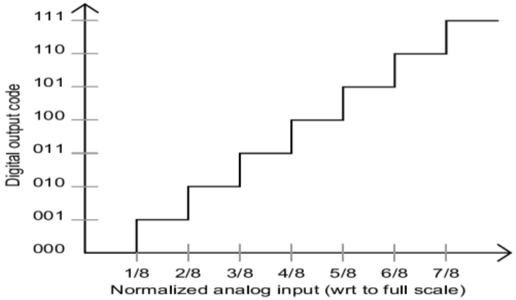


Fig : Uniform Quntization

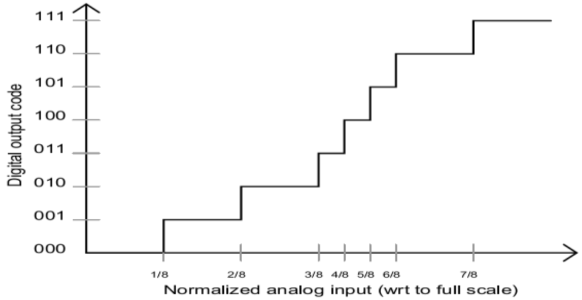


Fig : Non-uniform Quntization

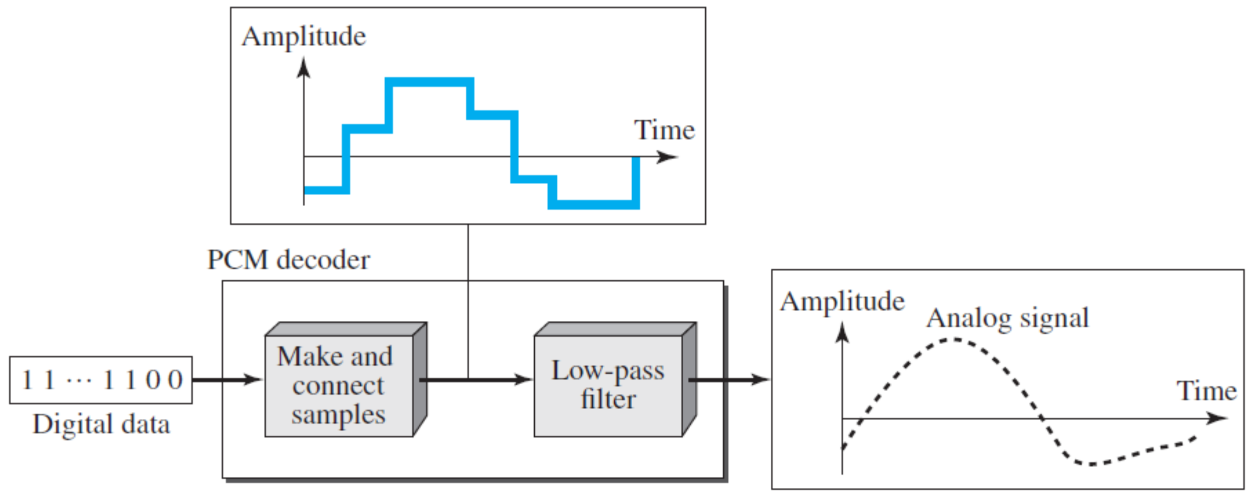
Original Signal Recovery (PCM Decoder)

To recover an analog signal from a digitized signal we follow the following steps:

- The decoder first uses circuitry to convert the code words into a pulse that holds the amplitude until the next pulse.
- After the staircase signal is completed, it is passed through a low-pass filter to smooth the staircase signal into an analog signal.
- The filter has the same cutoff frequency as the original signal at the sender.

The higher the value of L, the less distorted a signal is recovered.

Figure 4.27 *Components of a PCM decoder*



Bit rate and bandwidth requirements of PCM

- The bit rate of a PCM signal can be calculated

  Bit rate = n_b (number of bits per sample) x f_s (sampling rate)

- The bandwidth required to transmit this signal depends on the type of line encoding used.

- A digitized signal will always need more bandwidth than the original analog signal. Price we pay for robustness and other features of digital transmission.

Example 4.14

We want to digitize the human voice. What is the bit rate, assuming 8 bits per sample?

Solution

The human voice normally contains frequencies from 0 to 4000 Hz. So the sampling rate and bit rate are calculated as follows:

$$\boxed{\begin{array}{c}\text{Sampling rate} = 4000 \times 2 = 8000 \text{ samples/s} \\ \text{Bit rate} = 8000 \times 8 = 64{,}000 \text{ bps} = 64 \text{ kbps}\end{array}}$$

Maximum Freq = 4000 Hz
Fs = 2 x fmax
Bit rate = n_b x f_s

Error Detection- Introduction

- For most applications, a system must guarantee that the data received are identical to the data transmitted.

- Many factors (Attenuation - loss of energy, distortion - changes its form or shape, noise, and interference-unwanted signals) can alter one or more bits of a message. Some applications require a mechanism for detecting and correcting **errors.**

- Some applications can tolerate a small level of error. For example, random errors in audio or video transmissions may be tolerable, but when we transfer text, we expect a very high level of accuracy.

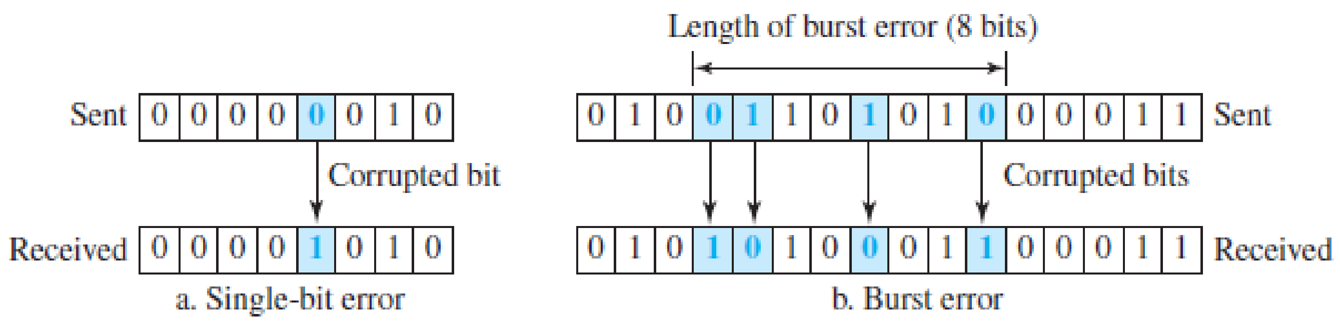
- Data link layer – error detection and correction functionality

Types of Errors

- Whenever bits flow from one point to another, they are subject to unpredictable changes because of **interference** (change the shape of the signal).

- ***Single-bit error:*** changed from 1 to 0 or from 0 to 1.
- ***Burst error:*** Two or more bits in the data unit have changed from 1 to 0 or from 0 to 1

Figure 10.1 *Single-bit and burst error*



Cyclic Redundancy Check (CRC)

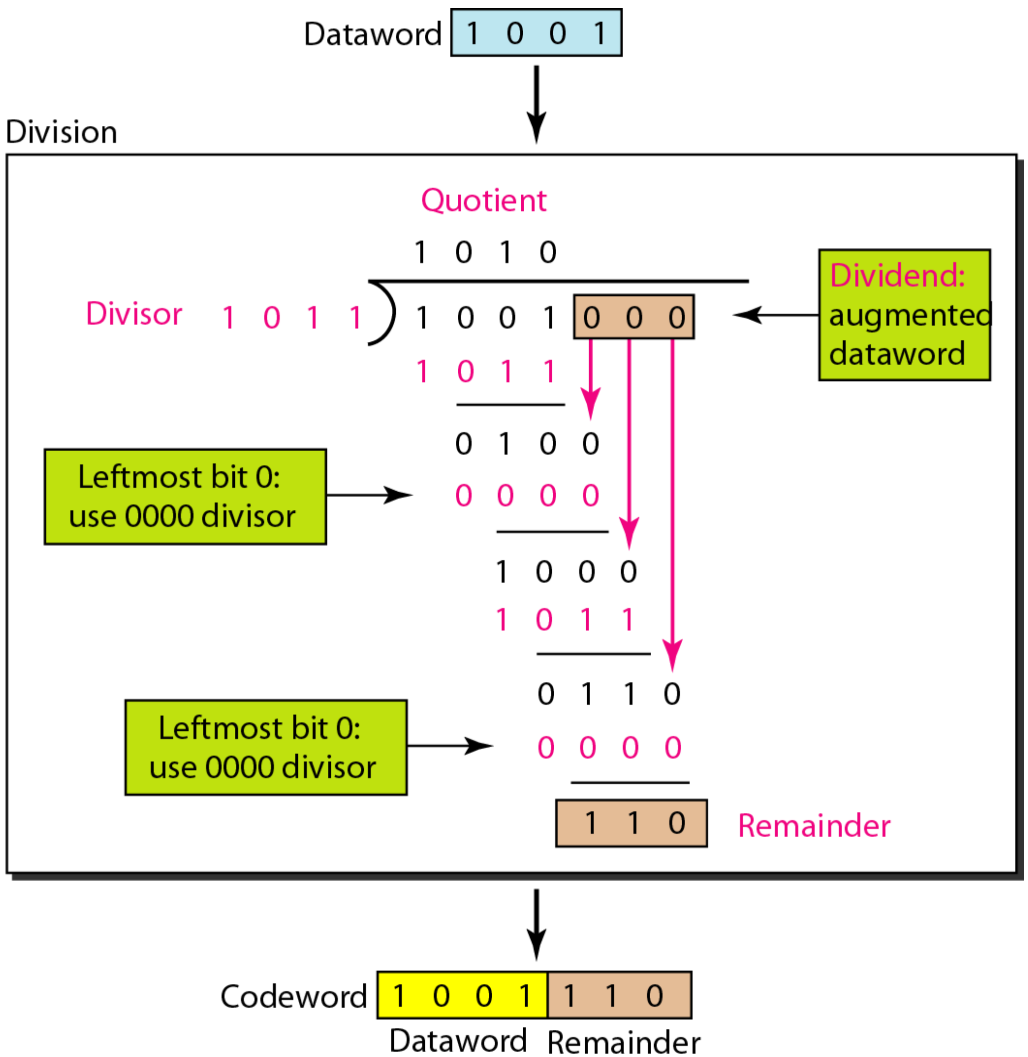
- Cyclic Redundancy Check (CRC) is an error detection method.
- It is based on binary division.
- The communicating parties agrees upon the size of message block and the CRC divisor.
- For example, the block chosen may be CRC (7, 4), where 7 is the total length of the block and 4 is the number of bits in the data segment. The divisor chosen may be 1011.
- Used in networks such as LANs and WANs

Table 10.3 *A CRC code with C(7, 4)*

Dataword	*Codeword*	*Dataword*	*Codeword*
0000	0000000	1000	1000101
0001	0001011	1001	1001110
0010	0010110	1010	1010011
0011	0011101	1011	1011000
0100	0100111	1100	1100010
0101	0101100	1101	1101001
0110	0110001	1110	1110100
0111	0111010	1111	1111111

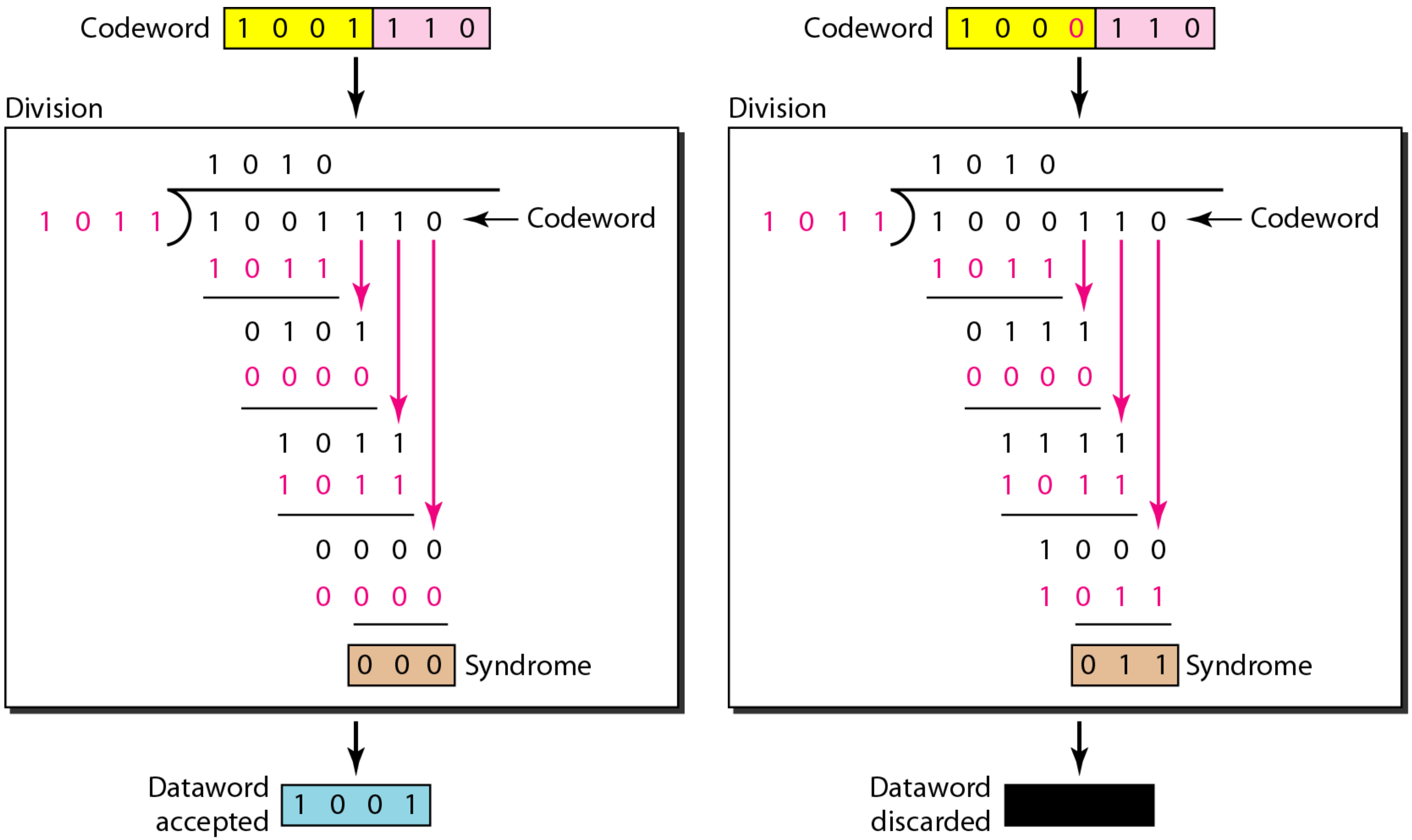
Figure 10.5 shows one possible design for the encoder and decoder.

Figure 10.15 *Division in CRC encoder*



Dataword:1001000
Divisor: 1011

Figure 10.16 ***Division in the CRC decoder for two cases***



Assume the data to be transmitted is
Data: 100100
Divisor:1101

Sender	**100100**	**000 (one number less than the divisor)**

```
           111101
1101    100100000
        1101
         1000
         1101
          1010
          1101
           1110
           1101
            0110
            0000
             1100
             1101
              001
```

Sender sends the codeword to receiver:
100100001

Assume the data to be transmitted is

Data: 100100

Divisor:1101

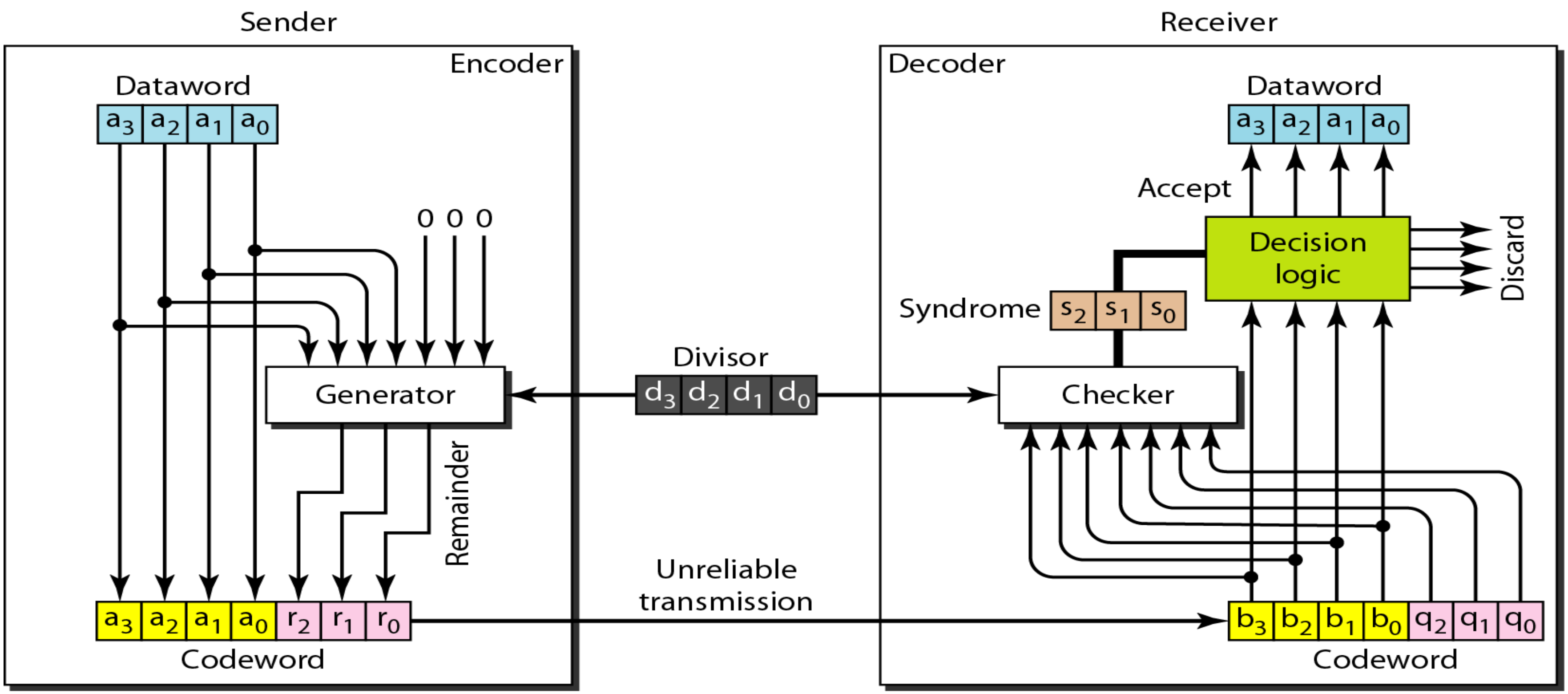
Receiver received $\boxed{100100001}$

```
           111101
1101       100100001
           1101
             1000
             1101
               1010
                1101
                 1110
                 1101
                   0110
                   0000
                     1101
                     1101
                     0000
```

0000

Syndrome=0 ---- no errors

CRC encoder and decoder



- The dataword has *k* bits (4 here); the codeword has *n* bits (7 here).
- The size of the dataword is augmented by adding *n* − *k* (3 here) 0s to the right hand side of the word. The *n*-bit result is fed into the generator.
- The generator uses a divisor of size n − k + 1 (4 here), predefined and agreed upon.
- The generator divides the augmented dataword by the divisor (modulo-2 division).
- The quotient of the division is discarded; the remainder (r2r1r0) is appended to the dataword to create the codeword.

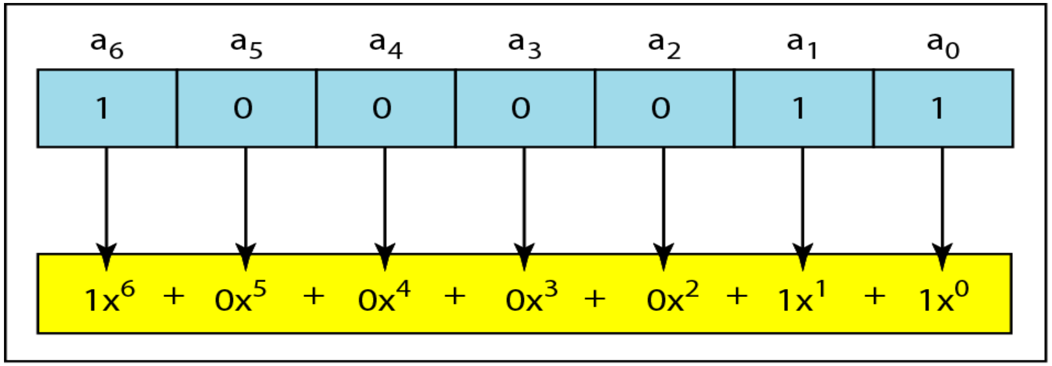
Question

A bit stream 10011101 is transmitted using the standard CRC method. The divisor is 1001.

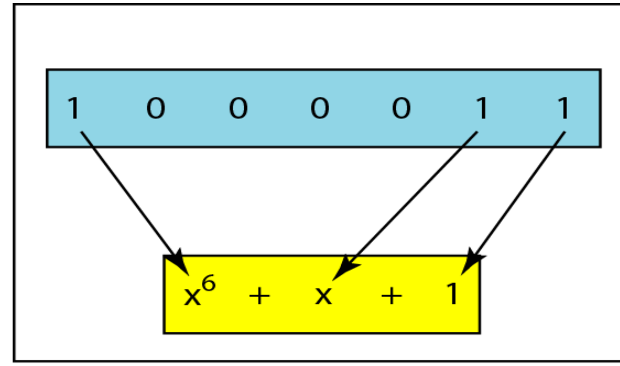
- What is the actual bit string transmitted?

Polynomials

- We can use a polynomial to represent a binary word.



a. Binary pattern and polynomial



b. Short form

Note

- The degree of a polynomial is the highest power in the polynomial. For example, the degree of the polynomial *x*6 + *x* + 1 is 6. Note that the degree of a polynomial is 1 less than the number of bits in the pattern.

Addition: x5 + x4 + x2
x6 + x4 + x2
x6 + x5

Multiplication: x3 × x4 is x7

Division: *x*6/*x*3, or *x*3

Polynomials

1. The information sequence (x) =100100000. Write the corresponding polynomial equation.

100100000 = **1. x^8**+0.x^7+0.x^6+**1.x^5**+0.x^4+0.x^3+0.x^2+0.x^1+0.x^0 = x^8 + x^5

2. What is the corresponding polynomial of i) 100 ii) 1101 iii) 1 0 1 0 1 0
100 – x^2
1101 – x^3+X^2+1
101010 - X^5 +X^3 +X
1101101 - X6+x5+x3+x2+1

The divisor in a cyclic code is normally called the *generator polynomial g(x)* or simply the *generator*.

Let Divisor / $g(x)=x^3 + x+1$. Consider the information sequence:1001000
$1001000=x^6 +X^3$

Sender

$$
\begin{array}{r|l}
 & X^3+x \\
\hline
x^3 + x+1 & x^6 +x^3 \\
 & \underline{x^6+x^4+x^3} \\
 & x^4 \\
 & \underline{x^4+x^2+x} \\
 & x^2+x
\end{array}
$$

Sender sends the codeword to receiver:
$x^6 +x^3+x^2+x$

Note: we continue to divide until the degree of the remainder is less than the degree of the divisor.

Let g(x)=x^3 + x+1. Consider the information sequence:1001011
1001011=x^6 +X^3

Receiver

X^3+x^1

x^3 + x+1 | x^6 +x^3+x^2+x

X^6+x^4+x^3

X^4+X^2+X

X^4+X2+X

000

Sender sends the codeword to receiver:

Question

- Generator: $x^3 + X^2 + 1$ Data : $x^8 + x^5$ Find the codeword

$X5+X4$

$x^3 + X^2 + 1$

$x^8 + x^5$

$X^8 + X^7 + X5$

X^7

$X^7 + x^5 + x4$

$x^5 + x4$

1

$x^3 + X^2 + 1$

$x^8 + x^5 + 1$

0

Standard polynomials

The divisor polynomials are such that it detects almost all odd bit errors and a few even bit errors. The commonly used polynomials in the literature are

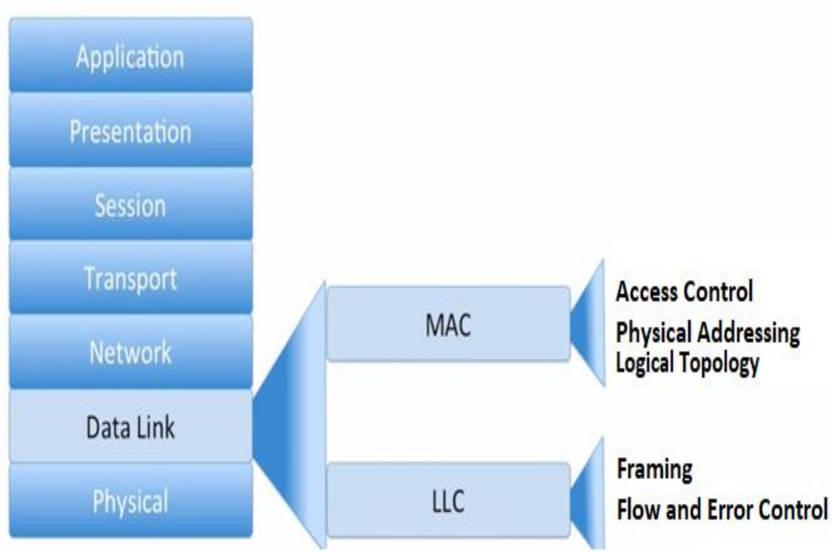
Table 10.4 *Standard polynomials*

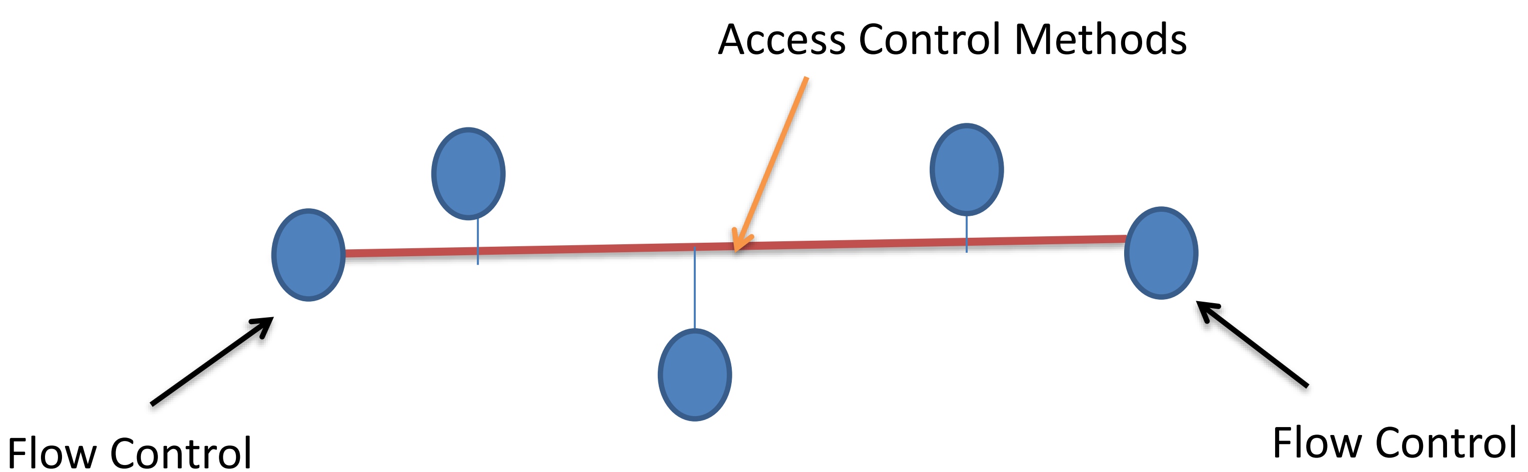
Name	*Polynomial*	*Used in*
CRC-8	$x^8 + x^2 + x + 1$ **100000111**	ATM header
CRC-10	$x^{10} + x^9 + x^5 + x^4 + x^2 + 1$ **11000110101**	ATM AAL
CRC-16	$x^{16} + x^{12} + x^5 + 1$ **10001000000100001**	HDLC
CRC-32	$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$ **100000100110000010001110110110111**	LANs

Note

- Asynchronous Transfer Mode and refers to a communication protocol which can be used to transfer data, videos and speech.
- ATM adoption layer is basically a software layer that accepts user data, which may be digitized voice, video or computer data, and makes them suitable for transmission over an ATM network.
- HDLC is a group of data link (Layer 2) protocols used to transmit synchronous data packets between point-to-point nodes.

Access Control

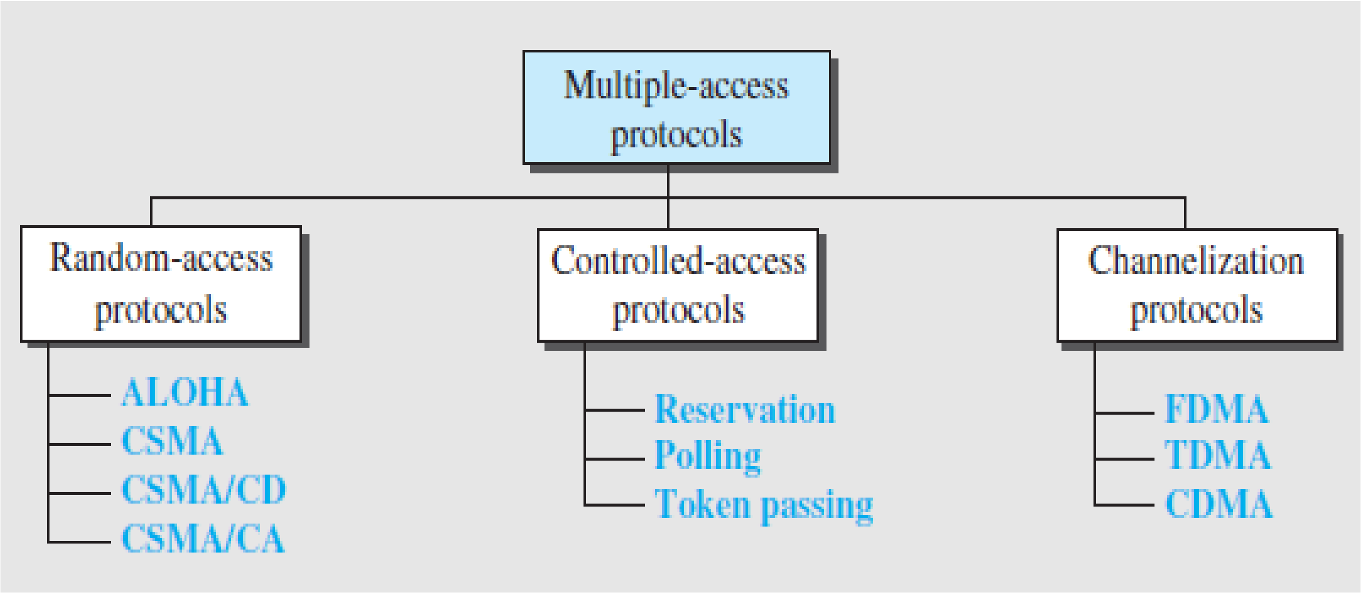




Multi Access Protocols

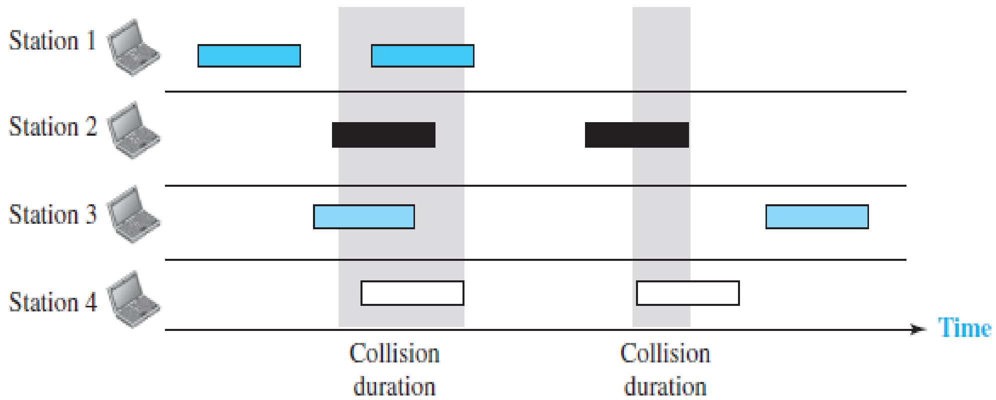
- When nodes or stations are connected and use a common link, called a multipoint or broadcast link, we need a multiple-access protocol to coordinate access to the link.

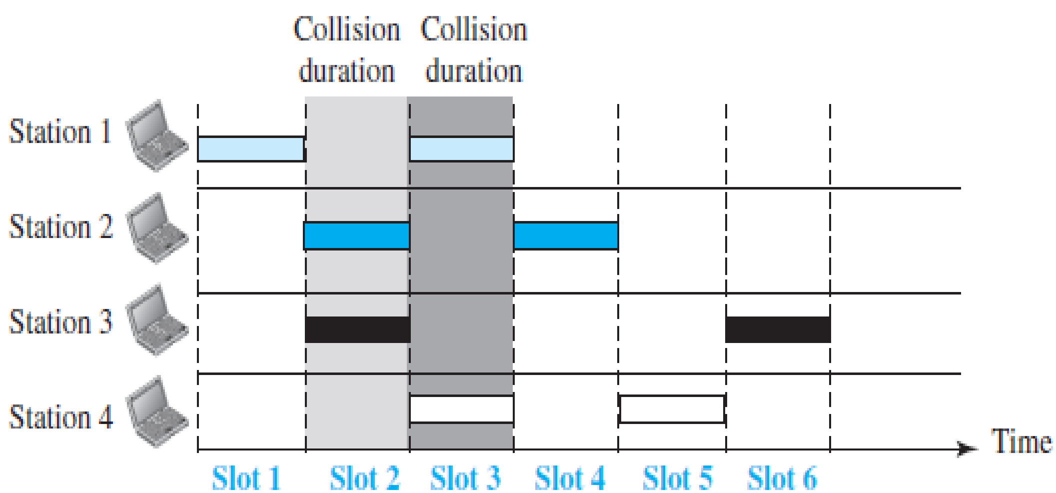
- To handle access to a shared link



Note (Pure Aloha and Slotted Aloha)

Figure 12.2 *Frames in a pure ALOHA network*





CSMA

- The chance of collision can be reduced if **a station senses the medium** before trying to use it.

- Principle “sense before transmit” or “listen before talk.”

- The possibility of collision still exists because of **propagation delay**, when a station sends a frame, it still takes time (although very short) for the first bit to reach every station and for every station to sense it.

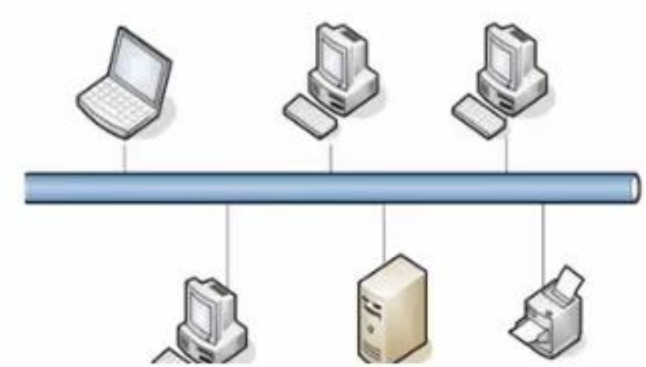
vulnerable time : time in which there is a possibility of collision.

Carrier Sense Multiple Access

Carrier means shared medium.
Sense means a node can listen and detect.

Multiple access:
Every node in the network has equal right to access to and use the shared medium, but they must take turns.

Carrier Sense Multiple Access

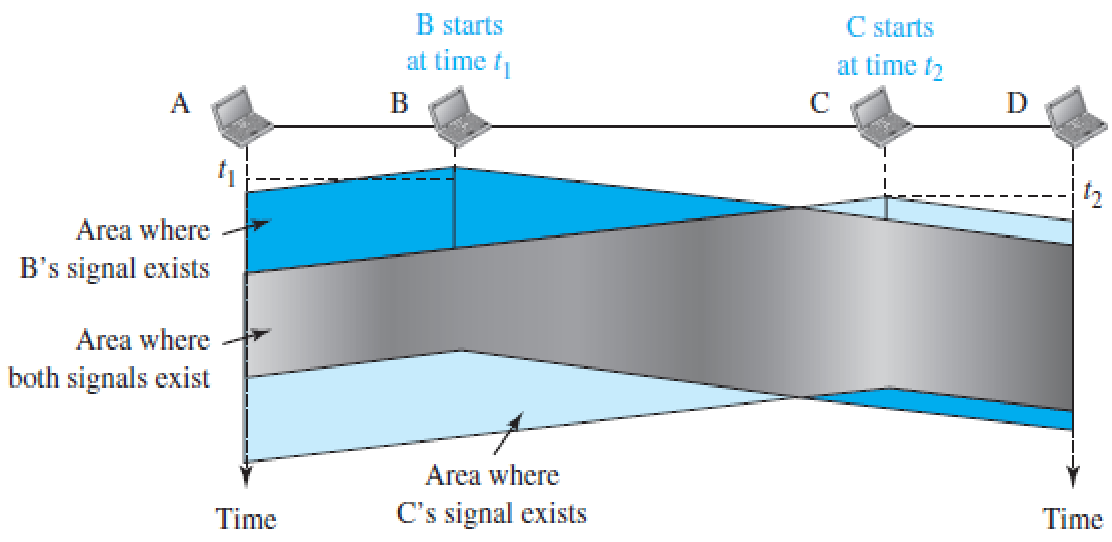


Carrier refers to transmission medium,
such as an electronic bus in Ethernet network,
or band of the electromagnetic spectrum
(channel) in Wi-Fi network.

when we say a node has a sense,
we really mean network interface card,
or NIC, has a sense.

The cable propagates the signal in both directions, so that the signal (eventually) reaches the NICs in all four of the computers.

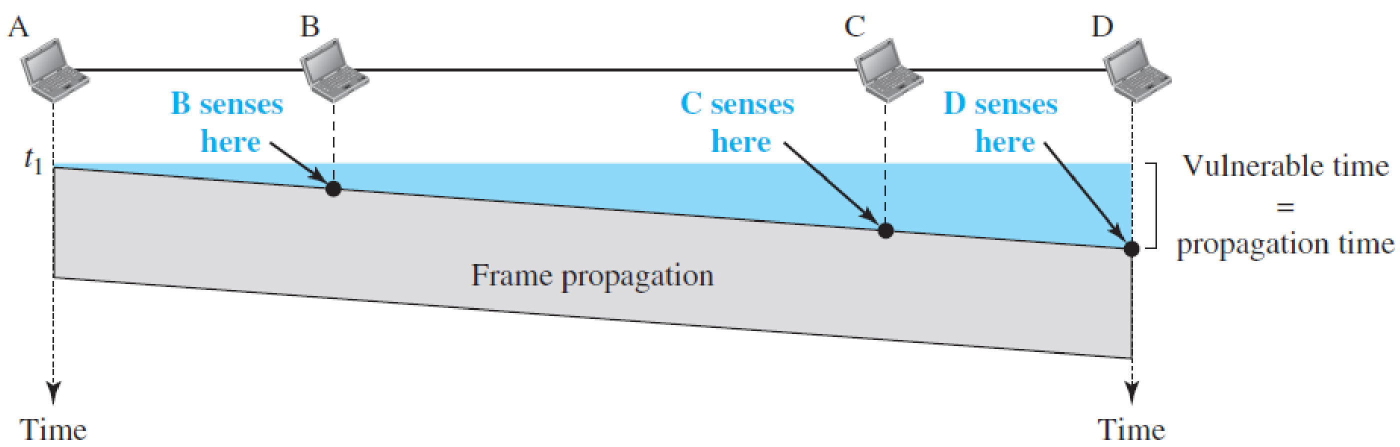
CSMA



- At time $t1$, station B senses the medium and finds it idle, so it sends a frame.
- At time $t2$ ($t2 > t1$), station C senses the medium and finds it idle because, at this time, the first bits from station B have not reached station C. Station C also sends a frame.
- Vulnerable Time= Propagation time (the time needed for a signal to propagate from one end of the medium to the other)

CSMA

Figure 12.8 *Vulnerable time in CSMA*

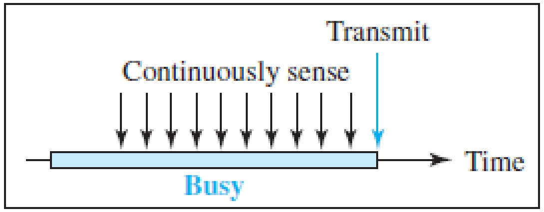


- The vulnerable time for CSMA is the ***propagation time*** *Tp*. This is the time needed for a signal to propagate from one end of the medium to the other.

Persistence Methods

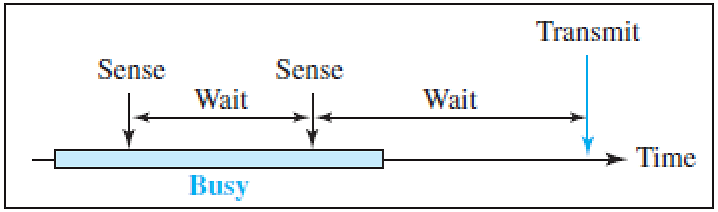
What should a station do if the channel is busy? What should a station do if the channel is idle?

- ***1-Persistent:*** After the station finds the line idle, it sends its frame immediately. This method has the highest chance of collision.



a. 1-Persistent

- ***Nonpersistent:*** A station that has a frame to send senses the line. If the line is idle, it sends immediately. If the line is not idle, it waits a random amount of time and then senses the line again. It reduces the chance of collision. it is ***unlikely that two or more stations will wait the same amount of time*** and retry to send simultaneously



b. Nonpersistent

Persistence Methods

P-Persistent

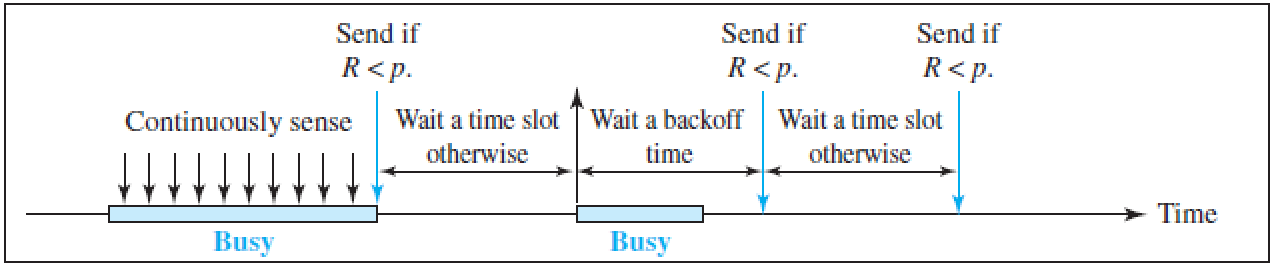
In this method, after the station finds the line idle it follows these steps:

1. With probability p, the station sends its frame.

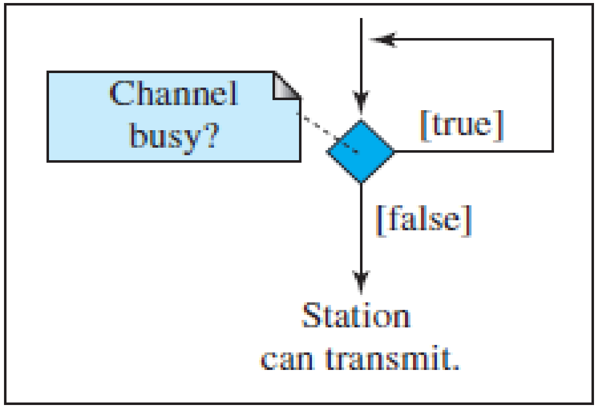
2. With probability $q = 1 - p$, the station waits for the beginning of the next time slot and checks the line again.

a. If the line is idle, it goes to step 1.

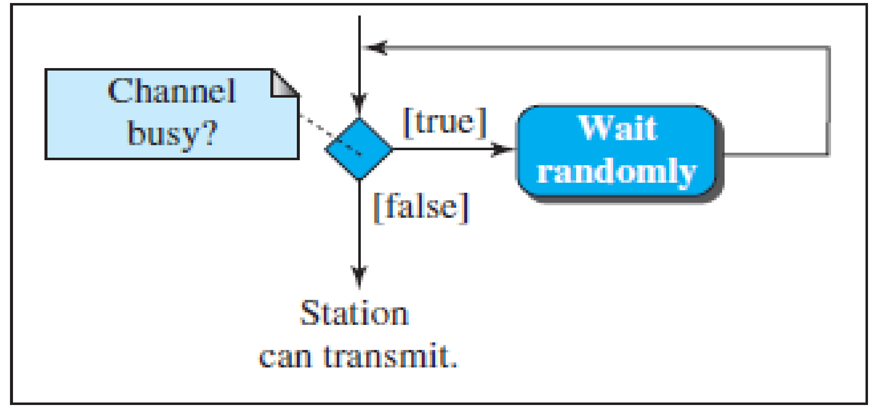
b. If the line is busy, it acts as though a collision has occurred and uses the backoff procedure.



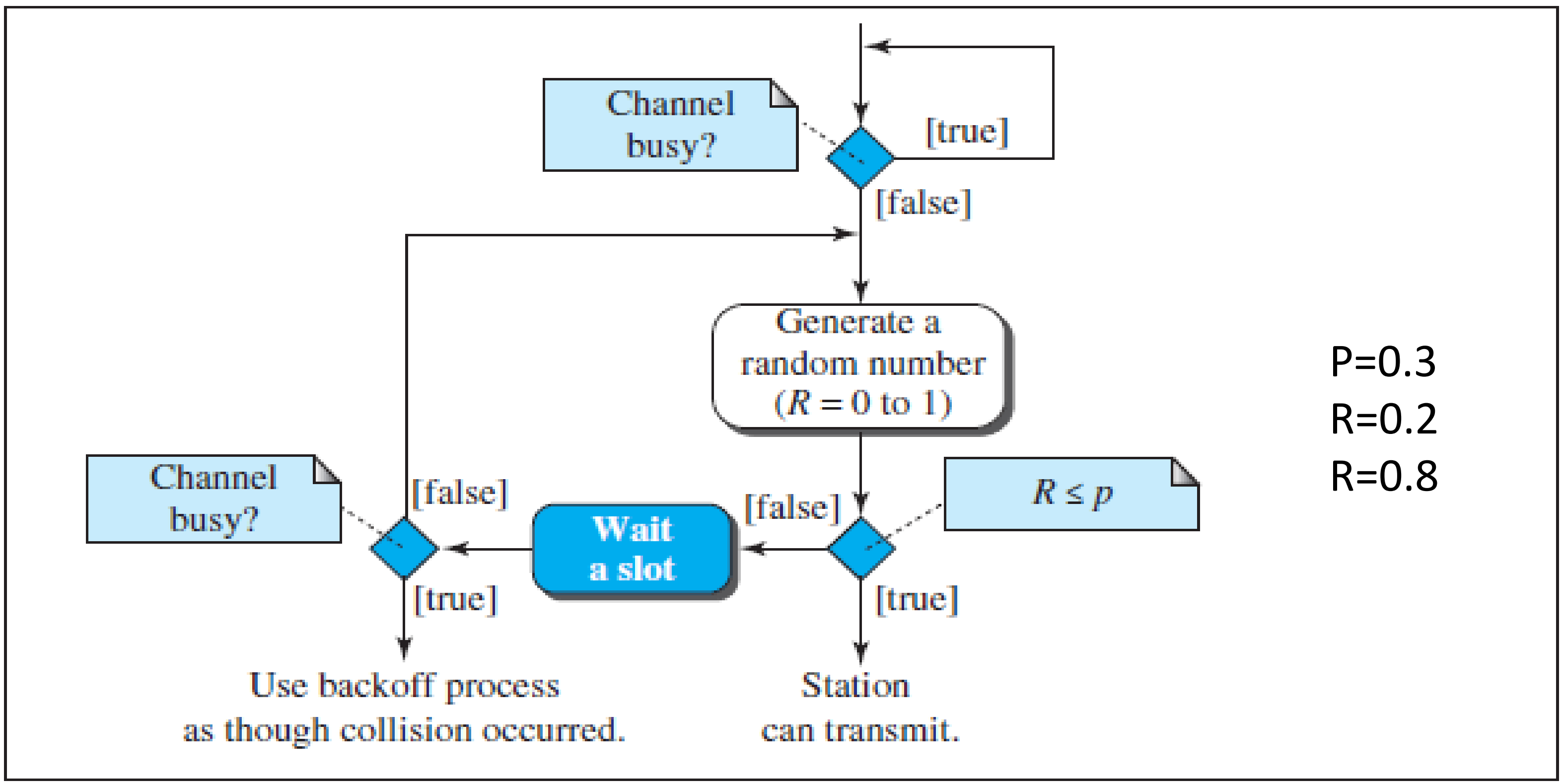
c. p-Persistent



a. 1-Persistent



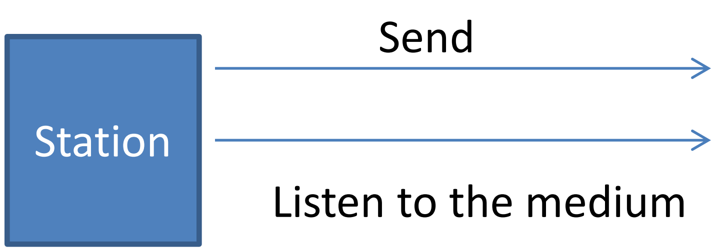
b. Nonpersistent



c. p-Persistent

CSMA/CD

- CSMA/CD protocol can be considered as a refinement over the CSMA scheme
- The nodes continue to monitor the channel while transmitting a packet and immediately cease transmission when collision is detected. This scheme is known as Carrier Sensed Multiple Access with Collision Detection (CSMA/CD) or Listen-While-Talk.



Principle: Transmission and collision detection are continuous processes

CSMA/CD

- It was used in Ethernet technology (Wired)
- We will not use Ack to detect the collision (No ACK).
- Uses collision signal

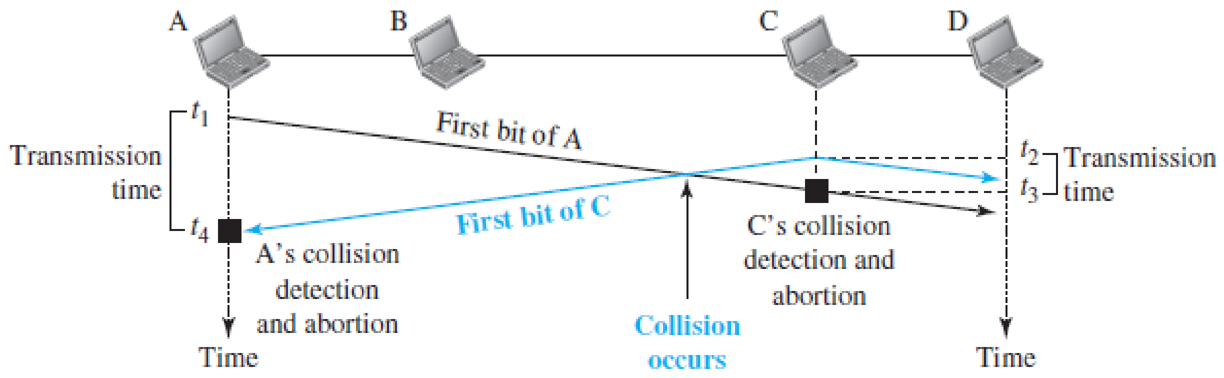
- A station **monitors** the medium after it sends a frame to see if the transmission was successful. If so, the station is finished. If, however, there is a collision, the frame is sent again.

1. We need to sense the channel before we start sending the frame by using one of the persistence processes (nonpersistent, 1-persistent, or p-persistent).

2. Transmission and collision detection are continuous processes.

CSMA/CD

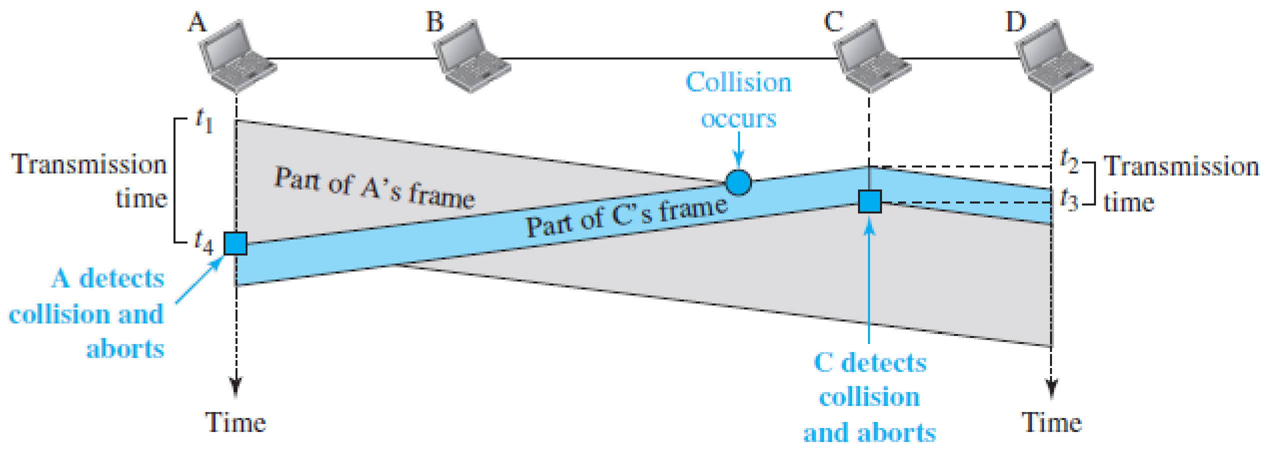
Figure 12.11 *Collision of the first bits in CSMA/CD*



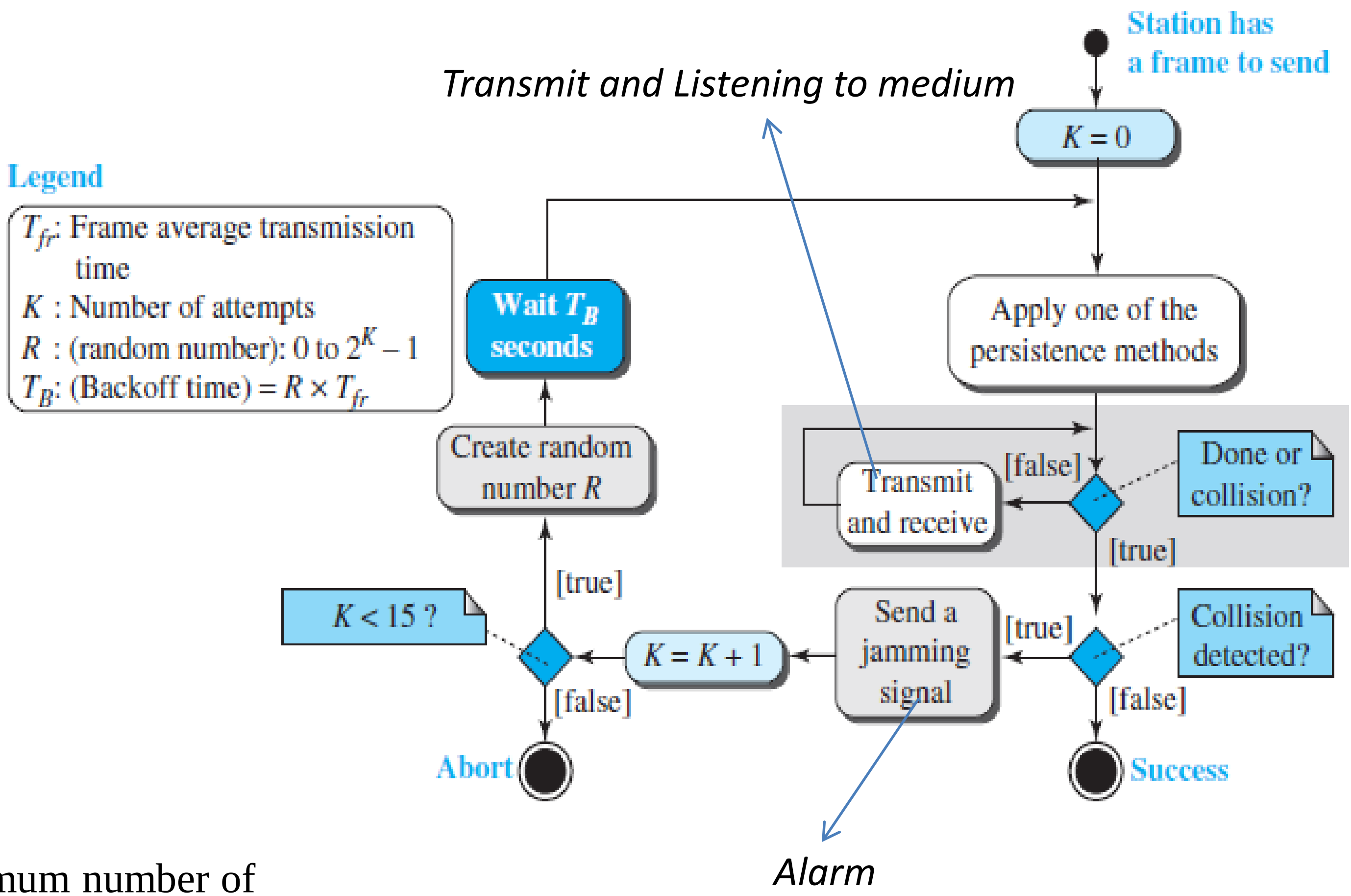
- At time $t1$, station A has executed its persistence procedure and starts sending the bits of its frame.
- At time $t2$, station C has not yet sensed the first bit sent by A.
- Station C executes its persistence procedure and starts sending the bits in its frame, which propagate both to the left and to the right.
- The collision occurs sometime after time $t2$.
- Station C detects a collision at time $t3$ when it receives the first bit of A's frame. Station C immediately aborts transmission.
- Station A detects collision at time $t4$ when it receives the first bit of C's frame; it also immediately aborts transmission

CSMA/CD

Figure 12.12 *Collision and abortion in CSMA/CD*



CSMA/CD



After a maximum number of retransmission attempts *Kmax*, a station must give up and try later

Throughput of CSMA/CD

- The maximum throughput occurs at a **different value of *G* and is based on the persistence method and the value of *p* in the *p*-persistent approach.**

G - the average number of frames generated by the system during one frame transmission time

- For the 1-persistent method, the maximum throughput is around 50 percent when $G = 1$.
- For the nonpersistent method, the maximum throughput can go up to 90 percent when G is between 3 and 8.

CSMA/CA

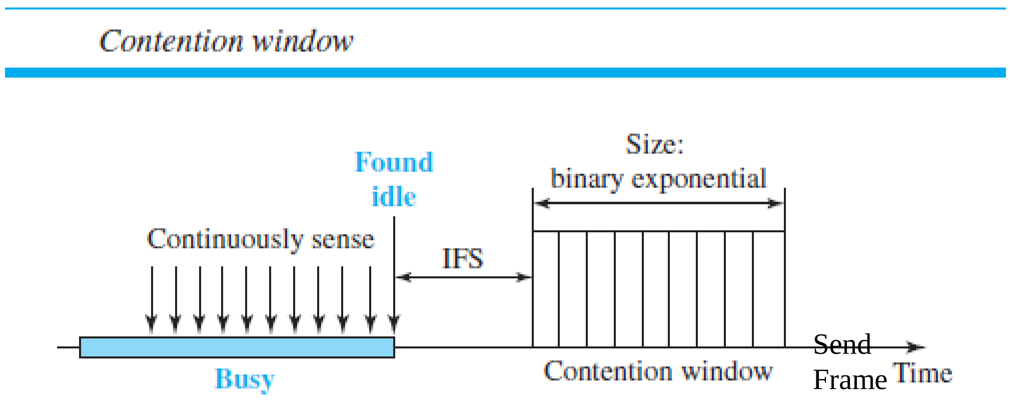
- Invented for wireless networks (wlan (IEEE 802.11) uses).
- Collisions are avoided through the use of CSMA/CA's three strategies:
  - the interframe space
  - the contention window
  - acknowledgments

Interframe Space (IFS):

- When an idle channel is found, the station does not send immediately. It waits for a period of time called the IFS.
- After waiting an IFS time, if the channel is still idle, the station can send, but it still needs to wait a time equal to the contention window.

CSMA/CA

- ***Contention Window:*** amount of time divided into slots. A station that is ready to send chooses a random number of slots as its wait time.
- The number of slots in the window changes according to the binary exponential backoff strategy. This means that it is set to one slot the first time and then doubles each time the station cannot detect an idle channel after the IFS time



- ***Acknowledgment:*** The positive acknowledgment and the time-out timer can help guarantee that the receiver has received the frame

CSMA/CA Flow Diagram

K=1 0-1
K=2 0-3
K=3 0-7
K=4 0-15

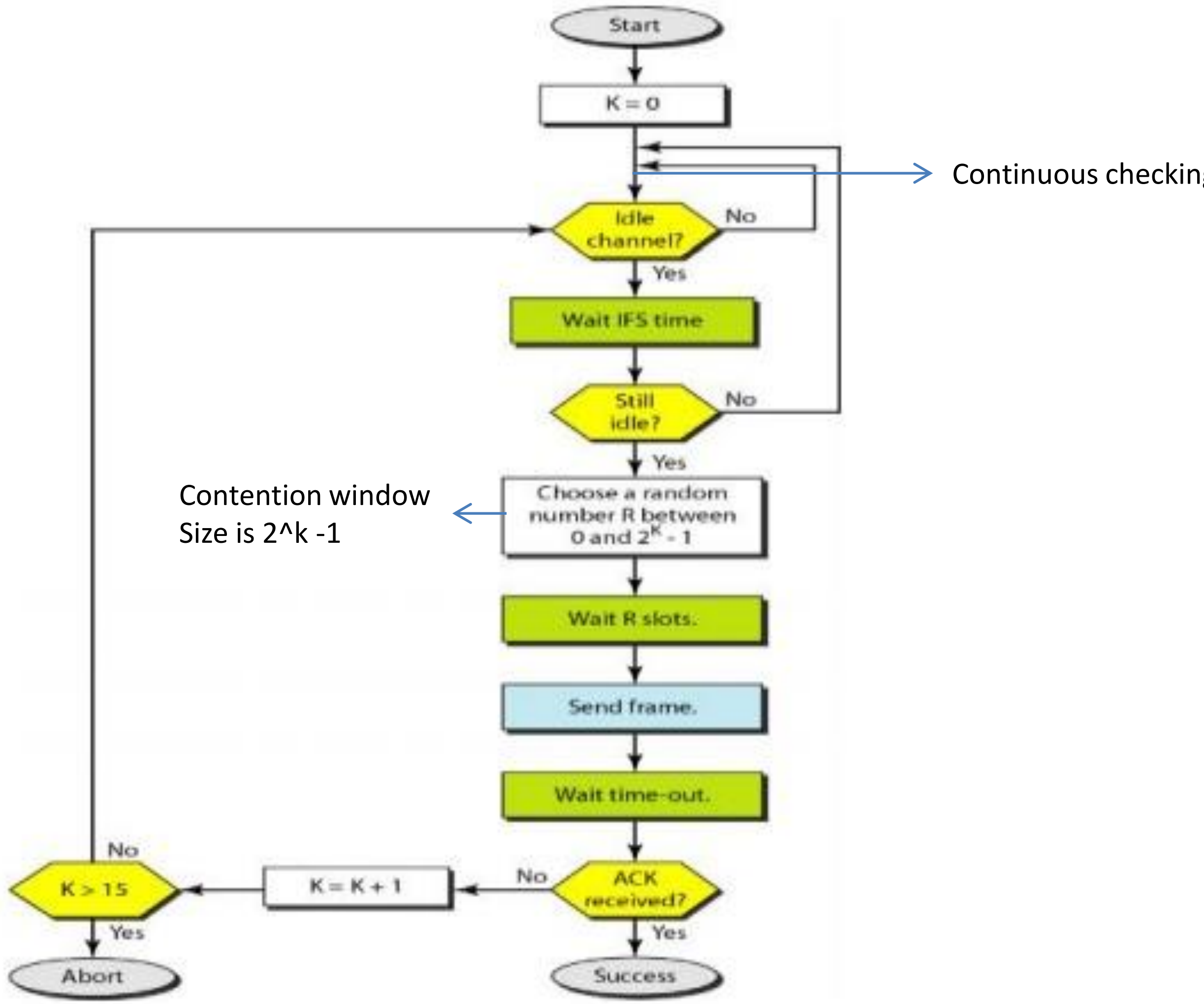


Figure 2.43 Flow diagram for CSMA/CA